

DURATION DEPENDENCE AND HETEROGENEITY: LEARNING FROM EARLY NOTICE OF LAYOFF*

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Abstract

This paper presents a novel approach to distinguish the impact of duration dependent forces and adverse selection on the exit rate from unemployment by leveraging variation in the length of layoff notices. I formulate a Mixed Hazard model in discrete time and specify the conditions under which variation in notice length enables the identification of structural duration dependence while allowing for arbitrary heterogeneity across workers. Utilizing data from the Displaced Worker Supplement (DWS), I employ the Generalized Method of Moments (GMM) to estimate the model. According to the estimates, the decline in the exit rate over the first 48 weeks of unemployment is largely due to the worsening composition of surviving jobseekers. Furthermore, I find that an individual's likelihood of exiting unemployment decreases initially, then increases until unemployment benefits run out, and remains steady thereafter. These findings are consistent with a standard search model where returns to search decline early in the spell.

Keywords: *Duration dependence, unemployment, job-finding rate, unobserved heterogeneity, hazard models, advance notice*

JEL Codes: *C41, E24, J64, J65*

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I INTRODUCTION

A well-established empirical regularity is that the exit rate from unemployment decreases over the unemployment spell, except for a spike at the time of unemployment insurance (UI) exhaustion. The decline in the exit rate may represent negative duration dependence, meaning that the longer a worker remains unemployed, the less likely they are to exit unemployment. This would be true if employers discriminate against long-term unemployed workers, or if workers lose valuable skills and connections over time that would otherwise help them find employment. However, workers with different unemployment durations, who may appear similar to researchers, may actually differ significantly. Factors such as employability, the urgency to find a job, or the ability to secure employment may vary across individuals. Such heterogeneity across workers would imply that the exit rate declines even in the absence of structural duration dependence. As more employable workers leave unemployment, the remaining pool of unemployed individuals increasingly consists of those who are less likely to exit unemployment.

Understanding how the likelihood of exiting unemployment evolves over the unemployment spell and the extent of heterogeneity across workers is crucial for the design of unemployment policies.¹ Furthermore, the magnitude and direction of structural duration dependence have implications for the incidence of long-term unemployment and the speed of recovery from economic downturns (Pissarides, 1992). Given its significance, a substantial body of literature has attempted to disentangle the sources of the decline in the exit rate from unemployment, with the Mixed Hazard (MH) model

¹See Shimer and Werning (2006), Pavoni and Violante (2007), Pavoni (2009), and Kolsrud et al. (2018).

([Lancaster, 1979](#)) serving as the standard framework. However, credible estimates have proven difficult to obtain.

In this paper, I develop and implement a novel approach to empirically disentangle the contributions of structural duration dependence and unobserved heterogeneity in explaining the dynamics of the exit rate from unemployment. My approach leverages variation in the length of notice workers receive from their employers before being laid off. Using data from the Displaced Worker Supplement (DWS), I compare workers with a notice period of more than 2 months (long notice) to those with a notice period of 1–2 months (short notice). I use inverse probability weighting (IPW) to achieve balance on a comprehensive set of observable characteristics across notice groups at the time of layoff. The analysis reveals that during the initial 12 weeks, the exit rate out of unemployment is 7 percentage points higher for long-notice workers. This difference is primarily due to a larger proportion of long-notice workers transitioning directly to their next job without experiencing unemployment. Strikingly, beyond the first 12 weeks, the exit rate for long-notice workers is actually lower.

I argue that the lower exit rate for long-notice workers at later durations is due to the composition of this group becoming relatively worse as a larger proportion of individuals exit early in the spell. This indicates heterogeneity among workers. In the presence of heterogeneity, such as differences in employability, more employable individuals exit unemployment earlier. As more workers exit the long-notice group early, the remaining workers in this group will have a lower proportion of highly employable workers than those in the short-notice group. Conversely, if there is no heterogeneity, a larger proportion of workers exiting early from the long-notice group will not alter the composition of long versus short-notice workers at later durations. Consequently, there would be no discernible difference in the exit rates of the two groups. Thus, the difference in the

exit rates of short and long-notice workers at later durations is indicative of the extent of underlying heterogeneity.² This is the fundamental idea behind my approach, which enables me to pin down the extent of heterogeneity and estimate structural duration dependence.

I operationalize this intuition by formulating a Mixed Hazard model in discrete time with multiple notice lengths. Within this framework, the probability of an individual exiting unemployment is given by the product of their unobservable type, a function of observable factors, and a structural hazard that varies with the duration of unemployment and the notice period. I show that structural duration dependence, characterized by how the structural hazard varies with the duration of unemployment, can be identified when two conditions hold. The first condition is conditional independence of notice length and the worker's unobservable type given observable characteristics (also referred to as unconfoundedness; [Rosenbaum and Rubin \(1983\)](#)). The second is a stationarity condition, which states that the length of the notice period does not affect the structural hazard at later unemployment durations.³

The identification utilizes exit rates weighted by the inverse of propensity scores. The weighted exit rates compare two hypothetical groups that differ only in the length of notice received but are otherwise comparable in both observable characteristics and, under conditional independence, unobserved types at the time of layoff. As a result,

²Moreover, the evidence suggests that most of this reflects unobserved rather than observable heterogeneity. As shown in Figure C6, the effect of observable composition changes over the unemployment spell on exit rates is small, and adjusting for them (via a Cox proportional hazards model) leaves the overall pattern of exit rates and the magnitude of the decline largely unchanged.

³The rationale for this assumption is that a longer notice affects job-finding through the additional search time it provides before separation. This extra time should matter primarily at the start of the spell, since for a worker who has already been unemployed for, say, three months or more, their job-finding probability may depend on unemployment duration through mechanisms such as employer discrimination or human capital depreciation, but should not depend on time spent searching before separation. Section III.B provides a search-theoretic rationale for this assumption and discusses potential violations.

differences in first-period exit rates identify the structural effect of notice. Because conditional independence rules out preexisting compositional differences and stationarity rules out a continuing direct effect of notice, differences in later-period exit rates reflect only the dynamic selection generated by the initial difference in exit rates. This identifies the extent of heterogeneity through the moments of the distribution of the composite type. The remaining variation in exit rates over duration then identifies the structural hazard. Building on the identification result, I develop a Generalized Method of Moments (GMM) estimator that also accommodates right-censored duration data.

I estimate the model using weighted moments from the DWS data. The estimates uncover substantial heterogeneity in individual exit probabilities. I find that a quarter of the 43.6% decline in the exit rate over the first five months is due to the changing composition of workers over the spell of unemployment. Moreover, after the first five months, an individual worker's exit probability increases until their unemployment benefit exhaustion and remains constant thereafter. This is in contrast to the exit rate, which continues to decline even after benefit exhaustion. I calibrate a partial equilibrium search model with a non-stationary environment ([Mortensen, 1986](#); [van den Berg, 1990](#)) and show that my findings can be rationalized in this framework with a decline in returns to search early in the spell.

Under the identifying assumptions in my framework, the lower exit rate among long-notice workers after the initial 12 weeks is attributed to heterogeneity. Two alternative explanations are possible. First, long- and short-notice workers could differ in their unobservables even after conditioning on observable characteristics. Second, a longer notice period could, in itself, lower exit probabilities at later durations. To assess both, in [Appendix E](#) I generalize the model to allow the distribution of unobserved heterogeneity to differ by a scale factor across notice groups, and to have the structural hazard

shift proportionally with notice length at later durations. These two deviations are not separately identified as the data provides only information about their combined effect. Requiring the implied heterogeneity distribution to be a valid distribution bounds this combined effect, and the bound lies entirely on the side that raises, rather than lowers, the estimated role of heterogeneity. The baseline estimates are therefore conservative with respect to both concerns: any deviation the data permit would reinforce, rather than overturn, the conclusion that the decline in exit rates reflects dynamic selection rather than genuine duration dependence.

This paper contributes to three strands of the literature.

First, the paper contributes to the literature on the identification and estimation of Mixed Hazard models. To the best of my knowledge, this is the first paper to establish a nonparametric identification result and develop a root- n consistent estimator under conditional, rather than unconditional, exogeneity of the identifying variable. Existing nonparametric identification results for single-spell data require a fully exogenous regressor, either time invariant and entering the hazard multiplicatively ([Elbers and Ridder, 1982](#); [Heckman and Singer, 1984a](#)) or time varying under a stationarity restriction ([Brinch, 2007](#)).⁴ In my application, the identifying variable is the length of advance notice, which, consistent with economic reasoning, is assumed to affect the structural hazard only at the beginning of the unemployment spell. However, the framework developed here can be readily adapted to other settings in which treatment is unconfounded and satisfies alternative restrictions on how its effect evolves over duration, including the proportional hazards specification.

Moreover, estimators based on the above-mentioned identification results converge

⁴An alternative route to identification relies on multiple spells ([Honoré, 1993](#)); however, this approach requires that the unobserved characteristics of the jobseeker remain constant across repeated spells.

at slower-than-parametric rates or require restrictive assumptions on the distribution of unobserved type to be feasible in practice.⁵ This is crucial because misspecification of unobserved heterogeneity can significantly impact estimates of structural duration dependence, as demonstrated by [Heckman and Singer \(1984b\)](#). In contrast, the discrete-time formulation of the model developed here delivers moment conditions that yield a tractable, root- n consistent GMM estimator that applied researchers can easily adapt to their own settings. To the best of my knowledge, [Alvarez et al. \(2021\)](#) is the only other study that utilizes moment conditions from a discrete version of the Mixed Hazard model and constructs a GMM estimator; however, their identification result and estimator pertain to multiple-spell data.⁶

Second, I contribute to the empirical literature on structural duration dependence. Previous empirical studies utilizing the Mixed Hazard model have had to make strong functional form assumptions due to challenges with estimation. Consequently, the evidence on structural duration dependence from these studies is mixed, as highlighted by [Machin and Manning \(1999\)](#) in their review. Recently, [Alvarez et al. \(2016\)](#) revived this strand of work by estimating a Mixed Hitting-Time (MHT) model ([Abbring, 2012](#)) using Austrian social security data. They focus on a selected sample of workers with multiple unemployment spells and are able to estimate the extent of heterogeneity across workers that is fixed between spells. A relative advantage of my approach is that it captures spell-specific heterogeneity.⁷ Another related study, [Mueller et al. \(2021\)](#), utilizes variation in expectations about job-finding from survey data to pin down variation in actual job-finding rates. While both of these studies also document substantial heterogeneity

⁵[Hausman and Woutersen \(2014\)](#) provides a succinct review of the literature.

⁶[van den Berg \(1990\)](#) also set up a discrete-time MPH model and study identification using cohort effects; however, they use a maximum likelihood estimator on aggregate data for estimation.

⁷For instance, a worker's savings or UI eligibility may change over the months or years by the time this worker becomes unemployed again.

across jobseekers, my estimator for structural duration dependence is flexible enough to capture changes around UI exhaustion.⁸

Finally, I contribute to the literature on the dynamics and determinants of job search. A substantial body of literature highlights a spike in exit rates at UI exhaustion, where exit rates increase until benefit exhaustion and decline thereafter.⁹ Standard search theory predicts an increase in an individual worker’s exit probability up to benefit exhaustion, consistent with the observed increase. However, it does not predict the subsequent decline, prompting behavioral modifications to the standard search theory such as storable job offers (Boone and van Ours, 2012) and reference dependence (DellaVigna et al., 2017). My estimates reproduce the increase in individual exit probabilities leading up to exhaustion, but show no decline after. The subsequent decline in the exit rate instead reflects the worsening composition of surviving workers, as a large share of workers exit at benefit exhaustion. I then show that a standard search model with heterogeneous workers can rationalize these patterns without any behavioral modifications.

II CONTEXT AND DATA

In this section, I describe the institutional setting and the data and document how the exit rate out of unemployment varies with the length of notice.

⁸Alvarez et al. (2016) utilizes an optimal-stopping model; a worker finds a job at an optimal stopping time when a Brownian motion with drift hits a barrier. Their model generates an inverse Gaussian distribution of duration for each worker. Mueller et al. (2021) restrict the structural hazard to be monotonic over the spell of unemployment, and their estimator yields a practically flat hazard.

⁹Katz and Meyer (1990) first documented the spike in exit rates at benefit exhaustion in the context of the US. Some recent papers that document this pattern using administrative data are DellaVigna et al. (2017) (Hungary), Ganong and Noel (2019) (US), and Marinescu and Skandalis (2021) (France).

II.A Institutional Details

Under certain circumstances, US employers are required to give notice of layoff. The federal WARN Act mandates that employers with 100 or more full-time employees provide a 60-day advance notice for plant closings and mass layoffs. A plant closing is defined as the shutdown of a site or units within it that results in 50 or more employees losing their jobs within a 30-day period, while a mass layoff is the loss of employment for 500 or more employees during a 30-day period, or 50-499 employees if they constitute one-third or more of the employer's active workforce. The law only applies to layoffs exceeding six months, excluding discharges for cause, voluntary departures, or retirements. Some states, such as California, New York, and Illinois, have implemented their own WARN laws that expand the coverage of employment losses beyond what the federal law requires.¹⁰

When it comes to unemployment insurance (UI), US workers who are terminated without cause are typically eligible to receive benefits for a limited duration. Although the UI program is a federal program, each state sets its own benefit levels and durations. Eligibility and benefits may depend on a combination of earnings, hours worked, or weeks worked during a base period, depending on the specific rules of the state's UI program. Typically, this base period consists of the first four out of five completed calendar quarters preceding the claim filing date. In most states, the maximum period for receiving benefits is 26 weeks. Nine states have a uniform benefit duration of 26 weeks, while the benefit durations in the remaining states vary depending on the applicant's earnings history. Additionally, a program for extended benefits has been in

¹⁰It is not possible to exploit policy variation across states, say in a differences-in-difference framework, due to confounding pre-trends; both California and New York implemented these laws in the aftermath of a national recession.

place since a 1970 amendment to the Federal Unemployment Tax Act (FUTA), which can be triggered by the state unemployment rate. Temporary programs have also been implemented to extend benefits during recessions.

II.B Data Description and Sample Construction

I use data from the Displaced Worker Supplement (DWS) for the years 1996-2020. DWS is fielded biennially along with the basic monthly Current Population Survey (CPS) in January or February. The survey is administered to individuals who report having *lost or left* a job within the past three years due to a plant closure, their position being abolished, or having insufficient work at their previous employment. In addition to information on workers' lost and current jobs, the DWS records both the length of advance notice received before layoff and the duration of subsequent unemployment, making it well-suited for the present application.

For the analysis, the sample is limited to individuals aged 21 to 64 years old. It excludes individuals who expected to be recalled to their previous jobs and those whose lost job was self-employment. To focus on workers who lost stable, permanent employment, the sample only includes individuals who were employed full-time for at least six months at their previous job and had health insurance benefits provided by that employer. The question on notice length in the DWS is categorical and includes categories of no notice, <1 month, 1-2 months, and >2 months. In my analysis, I focus specifically on the latter two categories.

While the majority of individuals in the DWS do not receive a notice at all, I exclude them from the sample because it is uncertain whether they were displaced or quit their

jobs voluntarily.¹¹ I also exclude individuals who received less than a month’s notice because they significantly differ from those with longer notice periods in observable characteristics. This is also reflected by a worse overlap in propensity scores when accounting for differences in observed characteristics while including this group, compared to when only the two longer notice period groups are included. It is possible that a notice of less than a month may refer to an informal verbal notice, which could be specific to certain jobs or industries. To ensure my findings are not confounded by differences in individual characteristics across notice groups, I exclude these individuals from the main sample. Nonetheless, results, including the <1 month notice group, as well as the no notice group, are presented in Section VB. Section C.1 in the Appendix provides additional details on data construction. Specifically, Table C1 summarizes the sample selection procedure, and Table C2 compares the characteristics of the workers in my sample to all workers in the CPS and DWS.

Throughout the remainder of the text, I use the term *long notice* interchangeably to refer to a notice period of more than two months and *short notice* to refer to a notice period of 1–2 months. Table 1 presents the summary statistics separately for workers with short and long notice in the sample. Columns (1) and (2) display the raw averages for the sample, revealing notable differences between the two groups. Workers with longer notice tend to be older and are more likely to be married. Additionally, workers laid off during plant closures are more likely to receive longer notice, potentially due to compliance with the WARN law. Workers with longer notice also tend to have longer job tenure. However, there are no notable differences in earnings for the two groups.

¹¹While the Displaced Worker Survey (DWS) seeks to capture separations caused by economic difficulties within firms, it is hard to distinguish layoffs from voluntary quits. This is because financially distressed firms might reduce hours or wages rather than lay off workers, leading some workers to voluntarily quit to seek better opportunities (Farber, 2017).

TABLE 1: DESCRIPTIVES BY NOTICE LENGTH

	Unbalanced			Balanced		
	Short (1)	Long (2)	Diff. (2)-(1)	Short (3)	Long (4)	Diff. (4)-(3)
Age	42.44 (0.24)	43.57 (0.22)	1.13*** (0.33)	43.04 (0.24)	43.01 (0.22)	-0.03 (0.33)
Female	0.45 (0.01)	0.46 (0.01)	0.02 (0.02)	0.46 (0.01)	0.46 (0.01)	-0.00 (0.02)
Married	0.59 (0.01)	0.63 (0.01)	0.04** (0.02)	0.61 (0.01)	0.61 (0.01)	-0.00 (0.02)
Black	0.10 (0.01)	0.09 (0.01)	-0.01 (0.01)	0.10 (0.01)	0.09 (0.01)	-0.00 (0.01)
College Degree	0.41 (0.01)	0.39 (0.01)	-0.03* (0.02)	0.40 (0.01)	0.40 (0.01)	-0.00 (0.02)
Plant Closure	0.46 (0.01)	0.62 (0.01)	0.16*** (0.02)	0.54 (0.01)	0.54 (0.01)	-0.00 (0.02)
Union Membership	0.15 (0.01)	0.16 (0.01)	0.01 (0.01)	0.15 (0.01)	0.15 (0.01)	0.00 (0.01)
In Metro Area	0.84 (0.01)	0.82 (0.01)	-0.01 (0.01)	0.83 (0.01)	0.83 (0.01)	0.00 (0.01)
Years of Tenure	7.12 (0.15)	9.18 (0.16)	2.06*** (0.22)	8.25 (0.16)	8.21 (0.15)	-0.04 (0.22)
Log Earnings	6.54 (0.01)	6.56 (0.01)	0.03 (0.02)	6.54 (0.01)	6.55 (0.01)	0.00 (0.02)
Observations	1959	2216		1959	2216	

Note: The sample consists of respondents from the Displaced Worker Supplement (DWS) for the years 1996-2020, who were between the ages of 21 to 64, who had worked full-time for at least six months at their previous job, received health insurance from their former employer, and did not expect to be recalled. Short notice refers to a notice period of 1-2 months, while long notice refers to a notice period exceeding two months. Columns (1) and (2) present raw averages for the sample, while columns (3) and (4) show weighted averages, where the weights correspond to the inverse of the estimated probabilities of receiving short or long notice.

To isolate the impact of notice from these additional correlates, which may affect the probability of exiting unemployment, I reweight the sample using IPW. I use a logistic regression model to predict the likelihood of receiving a longer notice based on several covariates. These covariates include age, gender, marital status, race, education, location characteristics, the reason for displacement, year of displacement, industry and occupation at the lost job, as well as union status, tenure, and earnings at the lost job. I then use the propensity scores to assign weights to the observations. Specifically, individuals with the long notice are assigned a weight of $1/\hat{p}(X_i)$, where $\hat{p}(X_i)$ is the estimated probability of receiving the long notice from the regression model for an individual with covariates X_i . On the other hand, individuals who received the short notice are assigned a weight of $1/(1 - \hat{p}(X_i))$. These weights adjust the composition of each notice group to match the distribution of observable characteristics in the full sample.

The summary statistics for the reweighted sample are presented in columns (3) and (4) of Table 1. After reweighting, the observable differences between the two groups disappear, indicating that the weights effectively minimize the observed disparities. Section C.2 in the Appendix provides additional details on propensity score estimation. Figure C1 demonstrates a high degree of overlap in the estimated propensity score distributions between long and short-notice workers. Additionally, Figures C2 and C3 depict the balance of the weighted sample with respect to the displacement year and industrial and occupational composition, respectively.

Even after balancing on observables, concern remains that workers with different notice lengths may differ in unobserved determinants of job finding, such as employability, labor market networks, or outside options. How important this concern is depends on how well the conditioning variables capture the determinants of notice assignment that are related to job-finding prospects. In the DWS data, separations stem from employer-

side economic conditions, leading to closures and downsizing. In these cases, how much notice firms provide is likely determined by institutional factors and the circumstances of the separation rather than individualized factors. Many of these factors, such as the nature of the separation, union status, seniority, job characteristics (proxied by occupation and education), and industry norms, are captured by the conditioning variables. A remaining concern is sorting across firms; for example, workers with higher unobserved job-finding ability may sort into larger firms with more formal HR practices. While these factors are not directly observed, the similarity of pre-displacement earnings and demographic characteristics across notice groups, even before reweighting, is somewhat reassuring regarding selection along these dimensions.

II.C Distribution of Unemployment Duration

In this section, I explore how a longer notice impacts the exit rate over the spell of unemployment. The exit rate at a given duration is defined as the share of individuals who found a job during that interval among those who were jobless at the beginning of the interval. Right-censored spells (individuals still unemployed at the time of the survey) are counted in the denominator for intervals up to their observed duration but not in the numerator. For the main estimates, unemployment duration is grouped into 12-week bins to reduce noise given the DWS's limited sample size.¹² Exits during the notice period (recorded as duration zero in the DWS) fall in the first 12-week bin.¹³ Figure 1 provides an overview of this setup.

¹²Sensitivity to the choice of bin size is examined in Section VB, and Figure C5 in the Appendix presents the data under alternative binning definitions.

¹³Treating the first 12 weeks as a single period makes the stationarity assumption less restrictive as notice length can affect the exit probability anywhere within the first 12 weeks rather than only before starting an unemployment spell, accommodating lingering effects or delays between finding a job and starting it.

FIGURE 1: OVERVIEW OF THE INSTITUTIONAL AND DATA SETUP



Note: The figure illustrates the timing of the notice period relative to the start of the unemployment spell and the duration bins used in the analysis. Shaded boxes represent the notice period. Workers who find a job during the notice period and never enter unemployment are recorded with an unemployment duration of zero in the DWS and are therefore included in the first bin (0–12 weeks). The dashed line marks the modal UI benefit exhaustion at week 26.

Workers who receive a layoff notice may start searching for a job before separating from their previous employer. In this case, some of these workers may secure a new job during the notice period, thus avoiding any period of unemployment. In the data, 14.7% of the workers with the short layoff notice report no duration of unemployment. Since workers with longer notice periods have more time to search for a new job while still employed, we expect their chances of avoiding unemployment to be even greater.

In Table 2, panel A, I examine the relationship between receiving a long notice and reporting an unemployment duration of 0. Columns (1) and (2) present estimates from unweighted regressions, while columns (3) and (4) present weighted regression estimates using the weights described in the previous section. Additionally, columns (2) and (4) include a comprehensive set of controls identical to the ones used to generate the weights. The table shows that the impact of a lengthier notice on the exit probability is reduced after accounting for observable characteristics of the separation. The coefficient in column (2) indicates that individuals who receive a longer notice are 8 percentage points more likely to avoid unemployment. Similar estimates are observed

TABLE 2: EXIT RATE – EARLY IN THE SPELL

	(1)	(2)	(3)	(4)
PANEL A. $\mathbb{I}\{\text{UNEMPLOYMENT DURATION} = 0 \text{ WEEKS}\}$				
> 2 month notice	0.094*** (0.012)	0.080*** (0.012)	0.077*** (0.013)	0.077*** (0.013)
PANEL B. $\mathbb{I}\{\text{UNEMPLOYMENT DURATION} \leq 12 \text{ WEEKS}\}$				
> 2 month notice	0.078*** (0.015)	0.074*** (0.016)	0.070*** (0.016)	0.070*** (0.016)
Controls	No	Yes	No	Yes
Weights	No	No	Yes	Yes
Observations	4175	4175	4175	4175

Note: The table presents estimates from linear regression models, where the main independent variable is an indicator variable that takes a value of 1 if the individual received a notice of more than 2 months and 0 if they received a notice of 1-2 months. The dependent variable is an indicator for reporting an unemployment duration of 0 weeks (Panel A) or less than 12 weeks (Panel B). The weights are generated using IPW. Robust standard errors are reported in parentheses.

in columns (3) and (4) as well. Notably, the inclusion of controls in column (4) does not lead to a change in the coefficient, indicating that the weighting has effectively achieved balance in terms of the covariates across the two groups. In panel B of Table 2, I present a similar regression analysis, but this time using an indicator for exiting unemployment within the first 12 weeks. The results show that the exit rate out of unemployment is around 7 percentage points higher for long-notice workers compared to the short-notice group.

Figure 2 presents the exit rate and the survival rate separately for the long- and short-notice workers over the spell of unemployment. Note that the rates are calculated using the weighted sample to ensure that the comparison is between groups of workers who

FIGURE 2: EXIT AND SURVIVAL RATE — LATER IN THE SPELL



Note: Short notice refers to a notice of 1-2 months, and long notice refers to a notice of more than 2 months. Panel A presents the weighted proportion of individuals exiting unemployment in each interval amongst those who were still unemployed at the beginning of the interval. Panel B presents the weighted proportion of individuals who are unemployed at the beginning of each interval. Error bars represent 90% confidence intervals.

are similar at the time of layoff but received different lengths of notice.¹⁴ Approximately 56.6% of individuals with a long notice exit within the first 12 weeks, while 48.7% of those with a short notice do the same. However, over the course of unemployment, individuals with shorter notice periods catch up, resulting in almost identical survival rates for both groups by the 48th week of unemployment. As shown in panel A of Figure 2, for all durations beyond 12 weeks, individuals with shorter notice periods have a higher exit rate compared to those with longer notice periods.

I interpret the higher exit rate for short-notice workers beyond the initial 12 weeks as evidence for heterogeneity across workers. When workers are heterogeneous, those

¹⁴Appendix D presents the unweighted exit rates and corresponding estimates obtained from the Mixed Hazard model. There is little qualitative difference between unweighted and weighted quantities, suggesting that differences in initial observable characteristics of the two groups are not driving the results.

with better chances of exiting unemployment do so earlier. Given that a larger proportion of long-notice workers exit earlier, the long-notice group will have a lower proportion of individuals with higher exit probabilities, which is reflected in the (average) exit rate. It is important to note that this interpretation holds under the condition that longer notice does not directly reduce the probability of exiting unemployment at later durations. In the following section, I formally outline the assumptions necessary to identify heterogeneity and duration dependence in a Mixed Hazard model, and I also discuss the plausibility of these assumptions and potential violations.

III ECONOMETRIC FRAMEWORK

This section shows how variation in notice length can be used to identify structural duration dependence. Section III.A sets up a discrete-time Mixed Hazard model with advance notice and describes the identification challenge. Section III.B introduces and discusses the two assumptions of conditional independence and stationarity, under which the key components of the model are identified. Section III.C presents the identification results together with the intuition behind them, and extends them to right-censored durations.

III.A *Mixed Hazard Model*

Prior to the layoff, each worker receives a notice period of length L , where L takes discrete values in the support $\mathcal{L}$. Workers are heterogeneous along two dimensions. First, each worker possesses an unobservable type ν . Second, workers differ in pre-notice observables X , which may include characteristics of the job, the worker, or the circumstances of the layoff. Using potential outcomes notation (Rubin, 1974), let $D_L \in \{1, 2, 3, \dots\}$ denote the potential unemployment duration under notice L . The econo-

metrician observes $D = \sum_{\ell \in \mathcal{L}} D_\ell \mathbb{I}\{L = \ell\}$.¹⁵ Hence, the potential duration under notice ℓ is observed only for workers who receive notice ℓ . For a set of conditioning variables Υ , define the potential hazard function $h_L(\cdot)$ and observed hazard function $h(\cdot)$ as follows:

$$h_L(d|\Upsilon) \equiv \Pr(D_L = d | D_L \geq d, \Upsilon), \quad h(d|\Upsilon) \equiv \Pr(D = d | D \geq d, \Upsilon).$$

Evaluated at $\Upsilon = (\nu, X)$, the potential hazard $h_L(d|\nu, X)$ is the probability that, under notice L , a worker of type ν with characteristics X exits unemployment at duration d , conditional on not having exited earlier. The Mixed Hazard structure specifies these individual exit probabilities $h_L(d|\nu, X)$ to be multiplicatively separable into three components: (i) a structural hazard $\psi_L(d)$ capturing structural duration dependence, (ii) an index $\phi(X)$ through which observable characteristics enter, and (iii) the unobserved type ν .¹⁶ The following assumption formalizes this structure.

Assumption 1 (Mixed Hazard). *For all $d \in \{1, 2, \dots\}$ and $\ell \in \mathcal{L}$,*

$$h_\ell(d|\nu, X) = \psi_\ell(d)\phi(X)\nu,$$

where, for some constant $\bar{\psi} > 0$, $\psi_\ell(d) \in (0, \bar{\psi}]$ and $\phi(X)\nu \in (0, 1/\bar{\psi}]$ almost surely.

It will prove useful to define the composite type $\theta(\nu, X) \equiv \phi(X)\nu$, since observed and unobserved heterogeneity enter the hazard only through their product. For brevity, I drop the arguments and write θ for this composite type throughout. The objective is to

¹⁵Throughout, $\mathbb{I}\{\cdot\}$ denotes the indicator function.

¹⁶Lancaster (1979) extended the proportional hazard model of Cox (1972) to incorporate unobserved heterogeneity. His Mixed Proportional Hazard (MPH) model expresses the hazard as the product of a structural hazard that varies with duration, a regression function, and the worker's unobserved type. The Mixed Hazard model here is a discrete-time counterpart of Lancaster's MPH model, except that notice is allowed to affect the hazard non-proportionally and singled out as the causal variable of interest, with the other observables in X treated as controls.

recover the structural duration dependence captured by $\psi_L(d)$, but this is complicated by the fact that the individual exit probabilities $h_L(d|\nu, X)$ are not observed. Instead, observed unemployment durations can be used to estimate the group-level exit rate by notice length, $h(d|L)$. Given the definition of the hazard function above and Assumption 1, this observed exit rate can be written as

$$h(d|L) = \mathbb{E}[h_L(d|\nu, X) | D \geq d, L] = \psi_L(d)\mathbb{E}[\theta | D \geq d, L].$$

Note first that the observed exit rate $h(d|L)$ differs from the potential exit rate $h_L(d)$, an instance of the fundamental problem of causal inference. The latter is the counterfactual exit rate that would prevail if the entire population were assigned notice length L , whereas $h(d|L)$ is the exit rate among workers actually assigned notice L . The usual approach from the treatment-effects literature to recovering such counterfactual quantities from observed data can be applied here. Yet even with the potential exit rate in hand, the challenge of disentangling structural duration dependence from dynamic selection remains. To see this, note that

$$h_L(d) = \psi_L(d)\mathbb{E}[\theta | D_L \geq d].$$

This formulation highlights that the potential exit rate reflects two forces: structural duration dependence, captured by $\psi_L(d)$, and dynamic selection, driven by changes in the average type among surviving workers, $\mathbb{E}[\theta | D_L \geq d]$. Moreover, the potential exit rate declines more steeply than the exit probability of an individual worker with a fixed type. This is because higher-type workers exit earlier, which decreases the average surviving type over the unemployment spell. Consequently, structural duration dependence and dynamic selection cannot be separately identified from the exit rates of a single notice group. This lack of identification is evident from the fact that a declining

exit rate is consistent with two polar cases: (i) negative structural duration dependence in the absence of heterogeneity, or (ii) constant or positive structural duration dependence combined with dynamic selection.

III.B Identifying Assumptions

In this section, I present and discuss the two key assumptions under which the model's components are identified. The consequences of relaxing these assumptions are examined in [Appendix E](#). The first is conditional independence, which states that among workers with similar observable characteristics, there is no systematic relationship between the length of notice received and the worker's unobservable type.

Assumption 2 (Conditional Independence). *The length of notice L is independent of the worker's unobservable type ν , conditional on observable characteristics X , i.e.,*

$$L \perp \nu \mid X.$$

In addition, overlap holds: every notice length has positive conditional probability, $p_\ell(X) \equiv \Pr(L = \ell \mid X) > 0$ almost surely for all $\ell \in \mathcal{L}$.

Note that this assumption does not require workers with different notice lengths to be similar on unobservables. Rather, it requires that any remaining differences in unobserved determinants of job finding not be systematically related to notice length after conditioning on observables. The credibility of this assumption, therefore, depends on how well the conditioning variables capture the drivers of notice length. As discussed in [Section II.B](#), the DWS setting provides some support for this restriction, as notice length there is likely set by institutional and circumstantial factors that the conditioning variables aim to capture.

The conditional independence assumption allows the potential exit rate $h_L(d)$ to be

recovered from the observed data using IPW. With the potential exit rates for more than one notice length in hand, identification is completed with the final assumption, stationarity, which states that the length of notice does not affect an individual's exit probability after the first period.

Assumption 3 (Stationarity). *For all L , X , and $d > 1$,*

$$\psi_L(d) = \psi(d).$$

The rationale for Assumption 3 is that workers with longer notice periods have more time to search for a new job before separating from their previous employer, which may increase their likelihood of finding a job at the start of their unemployment spell. However, suppose duration dependence in job-finding is caused by factors such as human capital depreciation due to prolonged unemployment or employers discriminating against long-term unemployed workers. In that case, a worker's exit probability later in the spell should only vary with the unemployment duration and not with the length of notice received at the onset of the spell. Given that I bin unemployment duration in 12-week intervals, this assumption implies that the length of notice affects only the probability of exit within the first 12 weeks, not after that.

More generally, Assumption 3 implies that individual exit probabilities vary only with the duration of unemployment and not with time elapsed since the start of the job search.¹⁷ This assumption would be violated if time spent searching increases or decreases an individual's likelihood of exiting unemployment. For instance, if workers learn while searching and become better at job search (Burdett and Vishwanath, 1988;

¹⁷This assumption aligns with a large class of search models, including those that involve non-stationarity. For instance, the model proposed by Lentz and Tranæs (2005), in which workers start searching harder over time as their savings run down, would be consistent with this assumption as savings only start depleting once unemployed.

Gonzalez and Shi, 2010), then those with longer notice would have a higher hazard even beyond the initial period. On the other hand, time spent searching may decrease the exit probability if workers first apply to all jobs in stock but subsequently only apply to newly posted jobs (Coles and Smith, 1998).¹⁸

III.C Identification Results

The identification argument proceeds through a sequence of results. Proposition 1 shows that conditional independence allows the potential exit rates for each notice group to be recovered from the data using IPW. Theorem 1 then shows that, under stationarity, the structural hazards and moments of the composite type distribution are identified from the potential exit rates of two notice groups. Proposition 2 and Corollary 1 extend these results to right-censored durations, as required for the application. Intuition for formal results is provided after their statement when appropriate. Proofs are presented in Appendix A.

Proposition 1. *Under Assumptions 1 and 2, for all $\ell \in \mathcal{L}$ and $d \geq 1$,*

$$h_\ell(d) = \mathbb{E} \left[\frac{\mathbb{I}\{L = \ell\} \mathbb{I}\{D = d\}}{p_\ell(X)} \right] / \mathbb{E} \left[\frac{\mathbb{I}\{L = \ell\} \mathbb{I}\{D \geq d\}}{p_\ell(X)} \right].$$

Recall that $p_\ell(X) = \Pr(L = \ell \mid X)$ is the propensity score. The idea behind IPW is as follows. In the observed data, workers with notice ℓ are not a random draw from the full population, since some covariate profiles X appear more often than they would if everyone were assigned notice ℓ . IPW addresses this by weighting each observation with $L = \ell$ by $1/p_\ell(X)$. This down-weights workers who were likely to receive ℓ and

¹⁸Another possibility for why individual exit probability may decline with time spent searching could be that workers get discouraged over time and stop trying. However, this seems unlikely since 85% of individuals in the sample eventually find employment after displacement, and among those who do not, none report being out of the labor force.

up-weights those who were unlikely to, so that the reweighted subsample matches the population distribution of X . Under Assumption 2, this also restores the population distribution of unobserved types ν , since L and ν are independent given X . As a result, the reweighted numerator and denominator correspond to exits and survivors at duration d in a population under universal assignment of notice ℓ , and their ratio recovers the potential exit rate $h_\ell(d)$.

Theorem 1. *Suppose Assumptions 1 and 3 hold. Then, for any $\ell, \ell' \in \mathcal{L}$ with $\psi_\ell(1) \neq \psi_{\ell'}(1)$ and any integer $\bar{D}$, the structural hazards $\{\psi_\ell(1), \psi_{\ell'}(1), \{\psi(d)\}_{d=2}^{\bar{D}}\}$ and the moments of the composite type distribution $\{\mathbb{E}[\theta^k]\}_{k=1}^{\bar{D}}$ are identified from the potential exit rates $\{h_\ell(d), h_{\ell'}(d)\}_{d=1}^{\bar{D}}$ up to a scale.*

Below, I present the intuition and sketch of the proof using the first two periods. In the first period, no one has exited yet, so the potential exit rate is given by

$$h_L(1) = \Pr(D_L = 1) = \psi_L(1)\mathbb{E}[\theta]. \quad (1)$$

In the second period, the exit rate is given by

$$h_L(2) = \frac{\Pr(D_L = 2)}{\Pr(D_L \geq 2)} = \frac{\mathbb{E}[\psi_L(2)\theta(1 - \psi_L(1)\theta)]}{1 - h_L(1)}.$$

Factoring out $\psi_L(2)$, expanding the expectation, using (1) to substitute $\psi_L(1) = h_L(1)/\mathbb{E}[\theta]$, and factoring out $\mathbb{E}[\theta]$ yields:

$$h_L(2) = \psi_L(2)\mathbb{E}[\theta] \cdot \underbrace{\frac{1 - h_L(1)\frac{\mathbb{E}[\theta^2]}{\mathbb{E}[\theta]^2}}{1 - h_L(1)}}_{\text{Compositional Adjustment}}. \quad (2)$$

Comparing (1) and (2) makes the identification problem clear. The first-period exit rate is proportional to the structural hazard since no compositional change has occurred yet. The second-period exit rate, in contrast, is distorted by the compositional adjustment

factor, which depends on $\mathbb{E}[\theta^2]/\mathbb{E}[\theta]^2$.

To see the role of this adjustment factor, consider two cases. When there is no unobserved heterogeneity ($\text{Var}(\theta) = 0$), $\mathbb{E}[\theta^2]/\mathbb{E}[\theta]^2$ equals one and the adjustment factor collapses to one. In this case, workers who exit first are no different from those who remain, so the exit rate ratio $h_L(2)/h_L(1)$ equals the structural hazard ratio $\psi_L(2)/\psi_L(1)$. When workers are heterogeneous, $\mathbb{E}[\theta^2]/\mathbb{E}[\theta]^2 > 1$ and the adjustment factor is below one. This is because high- θ workers exit at higher rates in the first period, worsening the composition of workers in the second period. As a result, the exit rate ratio falls below the structural hazard ratio, and this gap is larger the higher the heterogeneity. Therefore, if we knew $\mathbb{E}[\theta^2]/\mathbb{E}[\theta]^2$, we could recover the structural hazard ratio $\psi_L(2)/\psi_L(1)$ from the data.

Stationarity provides the missing restriction. By Assumption 3, the second-period structural hazard is common across notice lengths, $\psi_\ell(2) = \psi(2)$. Taking the ratio of (2) across two notice lengths ℓ and ℓ' , the common term $\psi(2)\mathbb{E}[\theta]$ cancels, leaving a single equation in the single unknown $\mathbb{E}[\theta^2]/\mathbb{E}[\theta]^2$:

$$\frac{h_\ell(2)}{h_{\ell'}(2)} = \frac{1 - h_\ell(1) \frac{\mathbb{E}[\theta^2]}{\mathbb{E}[\theta]^2}}{1 - h_\ell(1)} \bigg/ \frac{1 - h_{\ell'}(1) \frac{\mathbb{E}[\theta^2]}{\mathbb{E}[\theta]^2}}{1 - h_{\ell'}(1)}.$$

As long as the first-period exit rates $h_\ell(1)$ and $h_{\ell'}(1)$ differ across notice lengths, this equation uniquely pins down $\mathbb{E}[\theta^2]/\mathbb{E}[\theta]^2$, and hence the compositional adjustment factor and the structural hazard ratio $\psi_L(2)/\psi_L(1)$. A similar argument applies to identifying structural hazards beyond the second period, with each additional period identifying the next moment of θ and the corresponding structural hazard.¹⁹

¹⁹To understand why higher moments determine the hazard at later durations, we can consider how the composition of workers changes from $d = 2$ to $d = 3$. This change depends on the level of heterogeneity across workers at the start of $d = 2$. If the distribution of heterogeneity has a positive skew, the variance

So far, we have assumed that completed unemployment durations are observed. However, as is typical in many datasets, some individuals remain unemployed at the time of the DWS survey. For these individuals, we observe how long they have been unemployed, but we do not know if and when they will find a job. To formalize ideas, let D^c denote the elapsed time between the start of the unemployment spell and the survey. The econometrician observes $\tilde{D} = \min\{D, D^c\}$ together with a censoring indicator $C = \mathbb{I}\{D^c < D\}$, where D is the completed unemployment duration as before. The following proposition states that when D^c is random, the potential exit rate is still identified from the censored data. This assumption is plausible in the DWS, as the censoring time is determined by when an individual was surveyed, which should not correlate with individual outcomes.

Proposition 2. *Suppose Assumptions 1 and 2 hold and, in addition, $D^c \perp (L, v, X)$. Then, for all $\ell \in \mathcal{L}$ and $d \geq 1$,*

$$h_\ell(d) = \mathbb{E} \left[\frac{\mathbb{I}\{L = \ell\} \mathbb{I}\{\tilde{D} = d\} \mathbb{I}\{C = 0\}}{p_\ell(X)} \right] / \mathbb{E} \left[\frac{\mathbb{I}\{L = \ell\} \mathbb{I}\{\tilde{D} \geq d\}}{p_\ell(X)} \right].$$

Intuitively, for the numerator, note that $\mathbb{I}\{\tilde{D} = d\} \mathbb{I}\{C = 0\} = \mathbb{I}\{D = d\} \mathbb{I}\{D^c \geq d\}$, so it counts exits at duration d within the group whose censoring time exceeds d . Similarly, for the denominator, $\mathbb{I}\{\tilde{D} \geq d\} = \mathbb{I}\{D \geq d\} \mathbb{I}\{D^c \geq d\}$, so it counts survivals to d within the same group. Since D^c is random, averages over the restricted subsample therefore match those over the full population, and the potential exit rate is recovered as in Proposition 1. Combining Proposition 2 with Theorem 1 directly yields the following corollary, which is the main identification result utilized in the application.

among individuals who survive to $d = 2$ would be lower than that among individuals at the start of $d = 1$. This is because the few individuals with a high likelihood of exiting unemployment would have already left, reducing the variance among surviving workers.

Corollary 1. Suppose Assumptions 1–3 hold and $D^c \perp (L, \nu, X)$. Then, for any $\ell, \ell' \in \mathcal{L}$ with $\psi_\ell(1) \neq \psi_{\ell'}(1)$ and any integer $\bar{D}$, the structural hazards $\{\psi_\ell(1), \psi_{\ell'}(1), \{\psi(d)\}_{d=2}^{\bar{D}}\}$ and the moments $\{\mathbb{E}[\theta^k]\}_{k=1}^{\bar{D}}$ are identified up to scale from the joint distribution of the observed data $(\tilde{D}, C, L, X)$ up to a scale.

IV ESTIMATION

Generalized Method of Moments (GMM). Using the identification result presented in Corollary 1, we can use the Generalized Method of Moments (GMM) to construct a consistent estimator for the structural hazards and moments of the heterogeneity distribution. Since the model is identified only up to scale, I normalize the first moment by setting $\mathbb{E}[\theta] = 1$. With J possible notice lengths, the vector of unknown parameters is given by $\Theta = \{\{\psi_\ell(1)\}_{\ell=1}^J, \{\psi(d)\}_{d=2}^{\bar{D}}, \{\mathbb{E}[\theta^k]\}_{k=2}^{\bar{D}}\}$ and has a total of $2(\bar{D} - 1) + J$ unknown parameters.

Now, for each individual i , define the following moment condition:

$$m_i(\ell, d; \Theta) = \mathbb{I}\{L_i = \ell\} \cdot \left[\frac{\mathbb{I}\{\tilde{D}_i = d\} \mathbb{I}\{C_i = 0\}}{p_\ell(X_i)} - h_\ell(d; \Theta) \cdot \frac{\mathbb{I}\{\tilde{D}_i \geq d\}}{p_\ell(X_i)} \right]$$

Given the definition of $h_\ell(d; \Theta)$, it follows directly from Proposition 2 that $\mathbb{E}[m_i(\ell, d; \Theta)] = 0$ at the true parameter. We can now stack moment conditions pertaining to different notice lengths and durations in one vector, denoted by $\mathbf{m}_i(\Theta) = \{\{m_i(\ell, d, \Theta)\}_{d=1}^{\bar{D}}\}_{\ell=1}^J$. Observe that $\mathbf{m}_i(\Theta)$ contains $J \times \bar{D}$ moment conditions. As shown in Corollary 1, our parameters of interest are identified from these moment conditions as long as $J > 1$. This condition also ensures that the number of moments is equal to or greater than the number of parameters.

The corresponding sample average to $\mathbb{E}[m_i(\ell, d; \Theta)]$ can be written as:

$$\hat{m}(\ell, d; \Theta) = \frac{1}{n} \sum_{i=1}^n m_i(\ell, d; \Theta)$$

where n is the sample size. As before, stack the sample moments in a single vector $\hat{\mathbf{m}}(\Theta)$, such that $\hat{\mathbf{m}}(\Theta) = \sum_{i=1}^n \mathbf{m}_i(\Theta)/n$.

The GMM estimator $\hat{\Theta}$ is then given by: $\hat{\Theta} = \arg \min_{\Theta} \hat{\mathbf{m}}(\Theta)' \hat{W} \hat{\mathbf{m}}(\Theta)$. When the model is just-identified, $\hat{W}$ is given by the identity matrix. In the case of over-identification, the efficient weighting matrix is given by $\hat{W} = \hat{\Omega}^{-1}$, where $\hat{\Omega} = [\frac{1}{n} \sum_{i=1}^n \mathbf{m}_i(\hat{\Theta}) \mathbf{m}_i(\hat{\Theta})']$. Using the two-step estimation process, we can compute $\hat{\Theta}$. The asymptotic distribution of this estimator is given by $\sqrt{n}(\hat{\Theta} - \Theta) \rightarrow N(0, (\hat{M}' \hat{\Omega}^{-1} \hat{M})^{-1})$, where $\hat{M} = \partial \hat{\mathbf{m}}(\hat{\Theta}) / \partial \Theta$.

Functional Form for Structural Hazard. Even though the model is identified non-parametrically, given small sample sizes, to minimize the number of estimated parameters, I assume that the structural hazard $\psi(d)$ for $d > 1$ has a log-logistic form as follows

$$\psi(d) = \frac{(\alpha_2/\alpha_1)(d/\alpha_1)^{\alpha_2-1}}{1 + (d/\alpha_1)^{\alpha_2}} \quad (3)$$

where $\alpha_1 > 0, \alpha_2 > 0$. The hazard function in equation (3) is monotonically decreasing when $\alpha_2 \leq 1$ and is unimodal, initially increasing and subsequently decreasing when $\alpha_2 > 1$. The mode or the turning point is $\alpha_1(\alpha_2 - 1)^{1/\alpha_2}$.²⁰

²⁰This provides a flexible parametrization for the structural hazard relative to other commonly used parametrization, such as Weibull or Gompertz, as it allows the structural hazard to be non-monotonic. However, I also present non-parametric estimates in [Appendix D](#).

V DURATION DEPENDENCE AND HETEROGENEITY

VA *Baseline Estimates*

Table 3 presents the main estimates from the Mixed Hazard model. Since I normalized the mean of the worker type to 1 ($\mathbb{E}(\theta) = 1$), the estimated structural hazards for the first period for short- and long-notice groups coincide with their corresponding exit rates in the data. The last two lines in panel A of Table 3 show the estimated parameters for the log-logistic function specified in equation (3) used to model structural dependence.

The structural hazards implied by these parameters are presented in panel B of Table 3 and panel A of Figure 3. Additionally, panel A of Figure 3 shows the exit rate from the data, averaged across workers with short and long notice, alongside the estimated hazard. This figure shows that the estimated hazard consistently exceeds the empirical hazard throughout the unemployment spell, indicating the role of underlying heterogeneity. While the empirical hazard in the data declines by 43.6% over the first 24 weeks, the estimated structural hazard only decreases by 33.6% during the same period. Hence, after accounting for heterogeneity, almost three-quarters of the decline in the first 24 weeks can be attributed to structural duration dependence.

However, the estimated structural hazard increases by 75% from 12-24 to 24-36 weeks, a much more pronounced increase than the empirical hazard. This pattern possibly reflects individuals approaching the exhaustion of their unemployment insurance (UI) benefits.²¹ As previously noted, there is variation across individuals in the eligible duration of UI receipt. However, a significant proportion of individuals are eligible for

²¹Since the sample consists of displaced workers, a substantial portion of these individuals should be eligible for UI benefits. Table C4 in the Appendix shows that more than 80% of individuals unemployed for longer than 12 weeks report receiving UI benefits in the sample.

TABLE 3: ESTIMATION RESULTS

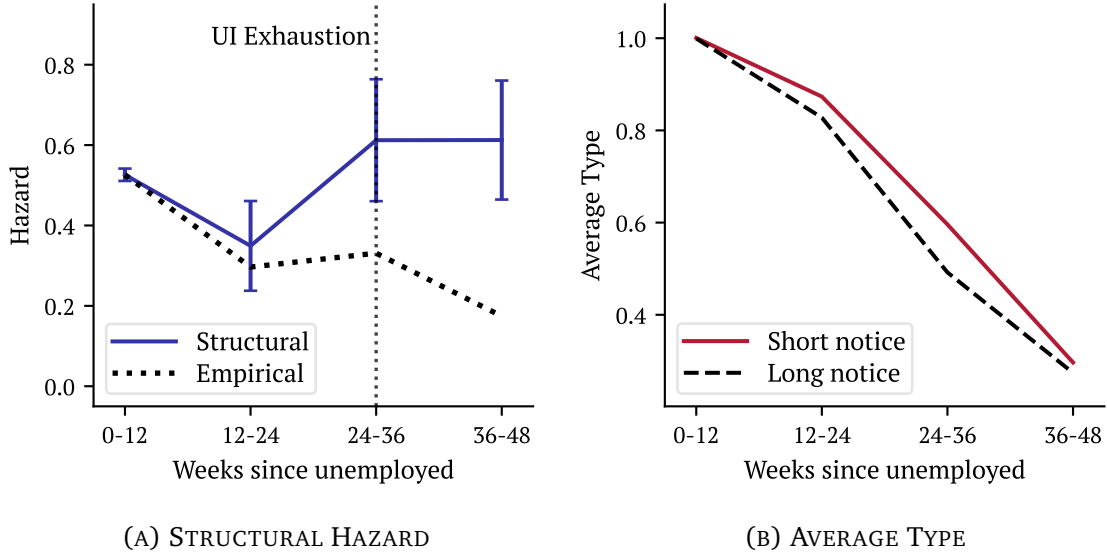
Parameter	Explanation	Estimate	SE
<i>Panel A: Estimated Parameters</i>			
$\psi_S(1)$	Structural hazard 0-12 weeks: Short notice	0.49	0.01
$\psi_L(1)$	Structural hazard 0-12 weeks: Long notice	0.56	0.01
α_1	Scale parameter for $\psi(d)$	2.06	0.17
α_2	Shape parameter for $\psi(d)$	2.54	0.27
<i>Panel B: Duration Dependence</i>			
$\bar{\psi}(1)$	Structural hazard: 0-12 weeks	0.53	0.01
$\psi(2)$	Structural hazard: 12-24 weeks	0.35	0.07
$\psi(3)$	Structural hazard: 24-36 weeks	0.61	0.09
$\psi(4)$	Structural hazard: 36-48 weeks	0.61	0.09
<i>Hansen-Sargan Test</i>			
Test statistic: 0.01		Critical value, $df = 1, \chi^2_{0.05}$: 3.84	

Note: The table presents estimates from the Mixed Hazard model. The first moment is normalized to one, and structural duration dependence is specified by equation (3). Panel A shows the estimated parameters from the model, and panel B presents structural hazards implied by the estimated parameters. The standard errors for the structural hazards are calculated using the delta method.

UI benefits that last for 26 weeks, which coincides with the third interval. Figure C4 in the Appendix shows that the proportion of individuals reporting exhausting their UI benefits jumps up significantly in the third interval. This finding of increasing structural hazard leading up to benefit exhaustion is consistent with individuals intensifying their job search efforts or lowering their expectations to secure employment before depleting their benefits. Finally, while the empirical hazard continues declining even after 36 weeks, the structural hazard remains constant.

Overall, the estimated pattern of structural duration dependence aligns with existing evidence from audit studies on call-back rates and with the predictions of search theory. I find that individual exit probabilities decline during the first five months, which can be attributed to duration-based employer discrimination. Further, I find that an individ-

FIGURE 3: BASELINE ESTIMATES



Note: The solid blue line in panel A presents estimates for structural hazards as implied by the estimated parameters in panel A of Table 3. The dotted black line in panel A presents the exit rate from the data, averaged across workers with short and long notice. Panel B presents the implied average type at each duration for those with short and long notice. Error bars represent 90% confidence intervals.

ual's exit probability increases leading up to benefit exhaustion and remains constant after. This is consistent with search theory, which predicts that individuals increase their search effort or lower their reservation wages until they reach benefit exhaustion. After that point, if there are no further changes in workers' incentives, their probability of exiting unemployment should remain constant. Interestingly, in their audit study, [Kroft et al. \(2013\)](#) find a decrease in callback rates only during the first six months of unemployment (refer to Figure 2 in their paper). In Section VI, I formally illustrate that my findings are consistent with a search model incorporating heterogeneous workers and falling callback rates early in the unemployment spell.

In recent studies, researchers have introduced behavioral modifications to the standard search theory in order to explain the observed decline in exit rates after UI exhaustion, which deviates from the predictions of the standard search model. Most notably,

[DellaVigna et al. \(2021\)](#) introduces reference dependence in utility to account for this decline. However, after adjusting for compositional effects, I do not find evidence of a decline in individual exit probabilities after UI exhaustion. Hence, I show that the data can be reconciled with the standard model by incorporating heterogeneous workers.

Panel B of Figure 3 plots $\mathbb{E}[\theta \mid D_\ell \geq d]$, the average type among all workers whose potential unemployment duration under notice length ℓ is at least d . The first interval (0–12 weeks) includes workers who found a job during the notice period and therefore report an unemployment duration of zero. So the first point represents the average type of the population at the time of layoff, $\mathbb{E}[\theta]$, which is the same under both notice lengths as no dynamic selection has yet occurred.²² From the second interval onward, dynamic selection sets in, and the average type declines under both notice lengths. Under long notice, the average falls faster because a larger share of high- θ workers have already exited during the first interval, so the two curves diverge between 12 and 24 weeks. By 36 weeks, once a substantial share of workers have exited under both notice lengths, the gap narrows again.

VB Robustness Checks

In this section, I examine the robustness of the main results by presenting non-parametric estimates, results using different moments for estimation, and a sensitivity analysis that jointly relaxes the two identifying assumptions and bounds the resulting bias.

²²It is also possible to obtain the group-specific average $\mathbb{E}[\theta \mid L]$ from the unweighted first-period exit rate using the estimated structural hazard, via the model relationship $h(1 \mid L) = \psi_L(1) \mathbb{E}[\theta \mid L]$. Doing so gives $\mathbb{E}[\theta \mid L = \text{long}] = 1.011$ and $\mathbb{E}[\theta \mid L = \text{short}] = 0.998$. The small difference between the two suggests that the observable characteristics associated with notice length do not translate into meaningful differences in average composite type at layoff, consistent with the close agreement between estimates using the weighted vs. unweighted exit rates, as discussed in Section [VB](#).

Appendix D presents results from several robustness checks. Figure D2 displays the data and estimates using the unweighted sample. The results reveal that the exit rates and estimates for the unweighted sample are very similar to those obtained using the weighted sample, suggesting that the observable characteristics play a limited role in explaining the differences in exit rates between notice groups. Figure D1 compares the non-parametric estimate for the structural hazard with the baseline log-logistic estimate. The non-parametric estimates are practically equivalent to the baseline but have larger standard errors, particularly for the last data point where the standard error increases significantly.

I also test the robustness of my findings by including additional notice length categories to the original estimation sample. First, I add individuals with no notice as a third group, and then I repeat the process by including individuals with <1 month notice as the third group. Details for these analyses are provided in Section D.3 in the Appendix, and estimates are presented in Figure D3. The results show that the estimated structural hazard is qualitatively similar to the baseline estimates in both cases. However, when adding the <1 month notice group, the increase in the structural hazard leading up to benefit exhaustion is less pronounced. Note that the model with three notice categories is overidentified with 4 degrees of freedom. The Sargan-Hansen J-statistic for testing overidentifying restrictions is reported in both cases and indicates no evidence against the null hypothesis of a correctly specified model.

For the main estimates, unemployment durations were grouped into 12-week intervals. I now assess the robustness of these findings to different bin sizes through two exercises. Firstly, I present estimates from the MH model using unemployment duration data binned into 9-week intervals. The estimated structural hazard, shown in Figure D4, mirrors the pattern reported in the main results. Specifically, it decreases

from the first to the second interval, followed by an increase in the third interval spanning 18-27 weeks, corresponding to UI exhaustion at 26 weeks. Secondly, I conduct a simulation where I calculate exit rates using various bin sizes under a specified data-generating process. Using these differently binned exit rates, I estimate the MH model to examine how the estimates are affected by the choice of bin size. Additional details for this simulation are presented in Section D.4 in the Appendix, and the results are illustrated in Figures D5 and D6. The takeaway is that the binned estimates reflect the cumulative structural hazard within an interval, representing the probability of an individual exiting unemployment at some point during that interval. Given this, while we may miss some intra-interval changes, the binned estimates still capture a meaningful quantity and are informative about the underlying duration dependence.

Finally, in Appendix E, I conduct a sensitivity analysis that allows for joint departures from the two identifying assumptions. To relax conditional independence, I allow the long-notice group's type distribution to differ from that of the short-notice group by a scale factor κ , where $\kappa > 1$ represents positive selection into long notice. To relax stationarity, I allow the long-notice group's structural hazard after the first period to differ from that of the short-notice group by a scale factor γ , where $\gamma > 1$ corresponds to a positive effect of longer notice on individual job-finding probabilities at later durations. The appendix first shows that, in this generalized model, the structural hazards and moments of the type distribution can be recovered as functions of $\eta = \gamma\kappa$, so the two deviations have identical implications for the estimates.

There are two main takeaways from this analysis. First, the direction and magnitude of the bias in the baseline estimates due to departures from these assumptions is governed by the value of η . Specifically, the recovered heterogeneity and structural hazards are monotonically increasing in η , so imposing $\eta = 1$ (baseline) understates both when

the true $\eta > 1$ and overstates them when $\eta < 1$.²³ Thus, the paper’s main conclusion, that a substantial share of the decline in exit rates reflects heterogeneity rather than genuine duration dependence, is threatened only if $\eta < 1$. Such a departure would arise either from negative selection into long notice ($\kappa < 1$) or from longer notice reducing the structural hazard later in the spell ($\gamma < 1$), or both. Second, requiring the recovered type distribution to be a valid distribution of positive types yields bounds on the composite parameter, and the entire admissible set for η lies above one. This suggests that the baseline estimates are conservative, since any deviation supported by the data would raise, rather than lower, the estimated structural hazards and the role of heterogeneity. Moreover, the qualitative pattern of structural duration dependence is identical to the baseline for all values of η within the set (see Figure E2).

VI A MODEL OF JOB-SEARCH

The estimates obtained from the Mixed Hazard model suggest a decline in an individual worker’s probability of exiting unemployment during the initial five months. Additionally, I find evidence that an individual’s likelihood of exiting unemployment increases as they approach the exhaustion of unemployment insurance (UI) benefits and remains constant after that. The latter is in contrast to the exit rate, which continues to decline even after benefit exhaustion. Researchers have tried to explain the decline in the exit rate after exhaustion using behavioral explanations such as storable

²³To see why, recall that under the baseline ($\eta = 1$), the entire difference between the long- and short-notice exit rates beyond the first period is attributed to dynamic selection. When $\eta < 1$, long-notice workers have lower individual job-finding probabilities beyond the first period, whether because of negative selection or a negative effect of notice. Consequently, less of the observed gap in exit rates needs to be attributed to compositional change, implying less unobserved heterogeneity than the baseline recovers. When $\eta > 1$, the reverse holds: the model requires more dynamic selection, and hence more heterogeneity, to reconcile the same observed pattern.

offers (Boone and van Ours, 2012) or reference dependence in utility (DellaVigna et al., 2021). In this section, I show that my findings align with standard search theory, incorporating heterogeneous workers, and are consistent with evidence from the audit study conducted by Kroft et al. (2013), which documents an initial decline in callback rates during the unemployment spell.

In particular, I set up a search model with heterogeneous workers. Within this model, workers choose search effort to maximize their expected utility. The likelihood of finding a job depends on the offer arrival rate and a worker’s search effort. Moreover, the offer arrival rate varies by the duration of unemployment and the type of worker. I calibrate the model to match the implied structural dependence to my estimate from the Mixed Hazard model and also match the exit rate implied by the model to the data. I then examine the trajectory of the offer arrival rate and search effort. This exercise also allows me to discern the impact on exit probabilities arising from two sources: the actions of optimizing agents in response to changing incentives and external factors directly influencing a worker’s employment prospects.

VI.A Model Setup

I consider a stylized model of job search where a worker’s search environment is non-stationary (Mortensen, 1986; van den Berg, 1990) and workers are heterogeneous. At every duration d , workers choose how much search effort s to exert to maximize their discounted expected utility.²⁴ Costs of search effort are given by the function $c(s)$, which is increasing, convex, and twice continuously differentiable, with $c(0) = 0$ and $c'(0) = 0$. The probability that a worker finds a job $\lambda(d, v, s)$ depends on the

²⁴Alternatively, the model could feature a reservation wage choice, and all conclusions about search effort would instead be regarding reservation wages.

time elapsed since unemployed d , their search effort s , and their type ν as follows: $\lambda(d, \nu, s) = \delta(d)\nu s$. Here, $\delta(d)\nu$ is the offer arrival rate, which varies over the duration of unemployment and across workers of different types. Once workers find a job, they remain employed forever. A worker receives unemployment insurance (UI) benefits $b(d)$ when unemployed and wages w when employed. The function $u(\cdot)$ gives the flow utility from consumption. Then the value function for a worker of type ν unemployed at duration d is given by:

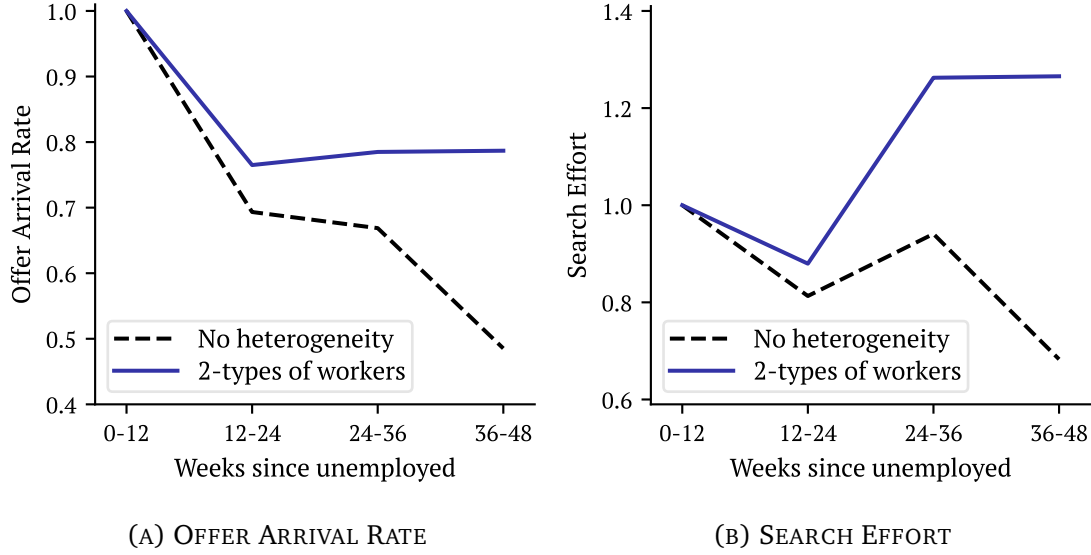
$$V_u(d, \nu) = \max_s [u(b(d)) - c(s) + \beta [\lambda(d, \nu, s)V_e + (1 - \lambda(d, \nu, s))V_u(d + 1, \nu)]$$

Here, β is the discount rate, and V_e is the value of employment given by $V_e = u(w) + \beta V_e$. The UI benefits $b(d)$ are equal to b for $d \leq D_B$ and equal to 0 otherwise. I also assume that after some time $D_T \geq D_B$ the job-finding function $\lambda(d, s, \nu)$ does not depend on the duration of unemployment d , such that for $d > D_T$, $\delta(d) = \delta_T$. This ensures that after D_T , jobseekers face a stationary environment, and hence, we can solve for the optimal search strategy of each worker in each period using backward induction. Finally, I consider the case of two types of workers: a high type H and a low type L with $\nu_H > \nu_L$, with π denoting the share of workers with the higher arrival rate.

VI.B Numerical Analysis

I now calibrate the model specified in the previous section. Let $s(d, \nu)$ denote a worker's optimal search effort at duration d . The probability that this worker finds a job $h(d|\nu)$ is given by $\delta(d)s(d, \nu)\nu$. So, a worker's exit rate evolves over the spell of unemployment due to changes in the offer arrival rate $\delta(d)$ and the worker's search effort. However, just as before, the exit rate $h(d) = \mathbb{E}[h(d|\nu)|D \geq d]$ also changes due to changes in composition over the spell of unemployment. I use my estimate

FIGURE 4: CALIBRATION OF THE SEARCH MODEL



Notes: The figure presents the search effort and the offer arrival rate from the calibration of the search model, assuming no heterogeneity (dashed black line) and assuming two types of workers (solid blue line). The search effort is averaged over two types of workers.

of the structural hazard from the Mixed Hazard model to target structural duration dependence $\mathbb{E}[h(d|\nu)]$ from the model.²⁵ Additionally, I match the exit rate implied by the model $h(d)$ to the data. In order to compare the predictions from this model to a model with no heterogeneity, I also calibrate the model assuming just one type of worker. In this case, I match the structural duration dependence or the exit rate implied by the model to the exit rate in the data. Further details for the calibration are provided in [Appendix B](#).

Figure B1 shows that both the model with and without heterogeneity fit the data almost perfectly. Figure 4 presents the search effort and the offer arrival rate implied

²⁵Note that the search model does not correspond precisely to the econometric framework since it does not imply that $s(d, \nu)$ evolves in the same manner for each type of worker. However, in [Appendix F](#), I simulate data from the search model with notice periods and show that my estimator does reasonably well in capturing movements in $\mathbb{E}[h(d|\nu)]$.

by the two calibration exercises. The offer arrival rate implied by the model with heterogeneity declines during the first five months and is constant after that, which is consistent with evidence from Kroft et al. (2013). Conversely, in the model with only one type of worker, the offer arrival rate continues to decline throughout the spell of unemployment. Finally, the model calibration implies that an individual's search effort decreases slightly during the first five months but then increases up to UI exhaustion and remains stable after that. In summary, the data and my findings can be rationalized with conventional search theory, without any behavioral adjustments, but by incorporating heterogeneity among workers and declining returns to search early in the unemployment spell.

VII CONCLUSION

In this paper, I develop a new approach to disentangle structural duration dependence from unobserved heterogeneity in the exit rate from unemployment, exploiting variation in the length of advance notice that workers receive prior to layoff. Under two identifying assumptions, conditional independence of notice and unobserved type and stationarity of the structural hazard beyond the initial period, I establish nonparametric identification of a discrete-time Mixed Hazard model and develop a root- n consistent GMM estimator. Beyond the advance-notice application, the framework can be adapted to other conditionally exogenous identifying variables whose effects on the structural hazard are concentrated near the start of the spell, such as severance pay. It also extends naturally to standard Mixed Proportional Hazard models in which the identifying variable proportionally shifts the hazard at all durations.

I estimate the model using data from the Displaced Worker Supplement (DWS) and

find substantial heterogeneity in individual exit probabilities. Roughly a quarter of the 43.6% decline in the exit rate over the first five months reflects the changing composition of surviving workers rather than a decline in individual job-finding probabilities. Moreover, after the first five months, an individual's exit probability rises up to UI exhaustion and remains constant thereafter, in contrast to the exit rate, which continues to decline. Researchers have suggested behavioral explanations for this post-exhaustion decline. I show that the alternative explanation of heterogeneity across individuals effectively accounts for this pattern. Specifically, as shown in the paper, my estimates can be rationalized within a standard search model with heterogeneous workers. These findings underscore the importance of incorporating heterogeneity when estimating and calibrating search models.

One limitation of the paper is that the identification strategy rests on the untestable assumption of unconfoundedness. While a sensitivity analysis in the paper explores the consequences of a violation and shows the qualitative conclusions remain intact, a design based on fully exogenous variation in notice length would nonetheless be more compelling. Firm-size thresholds in WARN coverage and tenure-based notice rules common in many European settings offer plausibly exogenous variation in notice, and verifying the findings of this paper in such settings would be a valuable direction for future research.

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APPENDIX A PROOFS AND DERIVATIONS

Proof of Proposition 1. Let Δ be any set of durations. Then,

$$\mathbb{E} \left[\frac{\mathbb{I}\{L = \ell\} \mathbb{I}\{D \in \Delta\}}{p_\ell(X)} \right] = \mathbb{E} \left[\frac{\mathbb{E} [\mathbb{I}\{L = \ell\} \mathbb{I}\{D_\ell \in \Delta\} | X]}{p_\ell(X)} \right] \quad (i)$$

$$= \mathbb{E} \left[\frac{p_\ell(X) \mathbb{E} [\mathbb{I}\{D_\ell \in \Delta\} | X]}{p_\ell(X)} \right] \quad (ii)$$

$$= \mathbb{E} [\Pr(D_\ell \in \Delta | X)] = \Pr(D_\ell \in \Delta) \quad (iii)$$

Step (i) applies the law of iterated expectations conditioning on X , with the weight passing outside the inner expectation, and replaces D with the potential duration D_ℓ , since $D = D_\ell$ for workers with $L = \ell$. Step (ii) uses Assumption 2: conditional on X , notice is independent of the worker's unobserved type and hence of the potential duration, so the inner expectation factors, with $\mathbb{E}[\mathbb{I}\{L = \ell\} | X] = p_\ell(X)$. Step (iii) cancels the propensity score, positive by the overlap condition, and applies the law of iterated expectations. Applying this identity to $\Delta = \{d\}$ and $\Delta = \{d, d + 1, \dots\}$ yields the numerator and denominator of the right-hand side expression in the statement of Proposition 1 as $\Pr(D_\ell = d)$ and $\Pr(D_\ell \geq d)$, respectively. Taking their ratio gives the potential exit rate $h_\ell(d)$.

Proof of Theorem 1. First note that the individual potential survival and density functions implied by Assumption 1 are

$$S_L(d | \theta) = \Pr(D_L > d | \theta) = \prod_{s=1}^d [1 - \psi_L(s)\theta], \quad S_L(0 | \theta) \equiv 1,$$

$$g_L(d | \theta) = \Pr(D_L = d | \theta) = \psi_L(d)\theta S_L(d - 1 | \theta),$$

Also, note the following relationships between the corresponding population survival, density, and hazard functions: $h_L(d) = g_L(d)/S_L(d - 1)$ and $g_L(d) = S_L(d - 1) - S_L(d)$. This implies that any one of the three determines the other two, so the argument below

uses whichever representation is most convenient.

Under Assumption 3, $\psi_L(d) = \psi(d)$ for $d > 1$, so we define

$$\check{S}(d | \theta) = \prod_{s=2}^d [1 - \psi(s)\theta] \quad \text{for } d \geq 2, \quad \check{S}(1 | \theta) \equiv 1.$$

Then, for $d \geq 2$,

$$S_L(d | \theta) = [1 - \psi_L(1)\theta] \check{S}(d | \theta), \quad (4)$$

and substituting (4) into the expression for the individual density gives

$$g_L(d | \theta) = \psi(d) [\theta \check{S}(d-1 | \theta) - \psi_L(1)\theta^2 \check{S}(d-1 | \theta)]. \quad (5)$$

Taking expectations of (4) and (5) yields, for $d \geq 2$,

$$S_L(d-1) = \mathbb{E}[\check{S}(d-1 | \theta)] - \psi_L(1)\mathbb{E}[\theta \check{S}(d-1 | \theta)], \quad (6)$$

$$g_L(d) = \psi(d) \{ \mathbb{E}[\theta \check{S}(d-1 | \theta)] - \psi_L(1)\mathbb{E}[\theta^2 \check{S}(d-1 | \theta)] \}. \quad (7)$$

Step 1: Identification of structural hazards. Consider two notice lengths $\ell, \ell' \in \mathcal{L}$ with $\psi_\ell(1) \neq \psi_{\ell'}(1)$, and take differences of (6) and (7) across ℓ and ℓ' :

$$S_{\ell'}(d-1) - S_\ell(d-1) = [\psi_\ell(1) - \psi_{\ell'}(1)] \mathbb{E}[\theta \check{S}(d-1 | \theta)], \quad (8)$$

$$g_{\ell'}(d) - g_\ell(d) = \psi(d) [\psi_\ell(1) - \psi_{\ell'}(1)] \mathbb{E}[\theta^2 \check{S}(d-1 | \theta)]. \quad (9)$$

From (8) and (9), we can write the two expectations as follows:

$$\mathbb{E}[\theta \check{S}(d-1 | \theta)] = \frac{S_{\ell'}(d-1) - S_\ell(d-1)}{\psi_\ell(1) - \psi_{\ell'}(1)}, \quad (10)$$

$$\mathbb{E}[\theta^2 \check{S}(d-1 | \theta)] = \frac{g_{\ell'}(d) - g_\ell(d)}{\psi(d) [\psi_\ell(1) - \psi_{\ell'}(1)]}. \quad (11)$$

Substituting (10) and (11) into (7) evaluated at $L = \ell$ and rearranging,

$$\psi(d) = \frac{\psi_\ell(1)g_{\ell'}(d) - \psi_{\ell'}(1)g_\ell(d)}{S_{\ell'}(d-1) - S_\ell(d-1)} \quad \text{for } d \geq 2, \quad (12)$$

where the denominator is non-zero by (8) since $\psi_\ell(1) \neq \psi_{\ell'}(1)$ and $\mathbb{E}[\theta \check{S}(d-1 \mid \theta)] > 0$.

Now note that for $d = 1$, $g_L(1) = \psi_L(1)\mathbb{E}[\theta]$, so plugging in $\psi_L(1) = g_L(1)/\mathbb{E}[\theta]$ in (12), we get:

$$\psi(d)\mathbb{E}[\theta] = \frac{g_\ell(1)g_{\ell'}(d) - g_{\ell'}(1)g_\ell(d)}{S_{\ell'}(d-1) - S_\ell(d-1)} \quad \text{for } d \geq 2.$$

Since the right-hand side is determined by $\{h_\ell(d), h_{\ell'}(d)\}_{d=1}^{\bar{D}}$, the structural hazards $\{\psi_\ell(1), \psi_{\ell'}(1), \{\psi(d)\}_{d=2}^{\bar{D}}\}$ are identified up to the common scale $\mathbb{E}[\theta]$.

Step 2: Identification of moments. Fix $L \in \{\ell, \ell'\}$ and expand $g_L(d \mid \theta) = \psi_L(d)\theta \prod_{s=1}^{d-1} [1 - \psi_L(s)\theta]$ for $d = 1, 2, 3, \dots$

$$\begin{aligned} g_L(1 \mid \theta) &= \psi_L(1)\theta, \\ g_L(2 \mid \theta) &= \psi_L(2)[\theta - \psi_L(1)\theta^2], \\ g_L(3 \mid \theta) &= \psi_L(3)[\theta - (\psi_L(1) + \psi_L(2))\theta^2 + \psi_L(1)\psi_L(2)\theta^3], \\ &\vdots \end{aligned}$$

Or, more compactly,

$$g_L(d \mid \theta) = \psi_L(d) \sum_{k=1}^{\bar{D}} c_k(d, \boldsymbol{\psi}_L) \theta^k, \quad (13)$$

where $\boldsymbol{\psi}_L = \{\psi_L(s)\}_{s=1}^{\bar{D}}$ and the coefficients satisfy $c_1(d, \boldsymbol{\psi}_L) = 1$ for all d , $c_k(d, \boldsymbol{\psi}_L) = c_k(d-1, \boldsymbol{\psi}_L) - \psi_L(d-1)c_{k-1}(d-1, \boldsymbol{\psi}_L)$ for $1 \leq k \leq d$, and $c_k(d, \boldsymbol{\psi}_L) = 0$ for $k > d$. Taking expectation of $g_L(d \mid \theta)$,

$$g_L(d) = \psi_L(d) \sum_{k=1}^{\bar{D}} c_k(d, \boldsymbol{\psi}_L) \mathbb{E}[\theta^k],$$

which in matrix form reads $\mathbf{g}_L = C(\boldsymbol{\psi}_L)\boldsymbol{\mu}$, where $\mathbf{g}_L = \{g_L(d)\}_{d=1}^{\bar{D}}$, $\boldsymbol{\mu} = \{\mathbb{E}[\theta^k]\}_{k=1}^{\bar{D}}$, and $C(\boldsymbol{\psi}_L)$ is the $\bar{D} \times \bar{D}$ upper-triangular matrix with entries $C_{d,k}(\boldsymbol{\psi}_L) = \psi_L(d)c_k(d, \boldsymbol{\psi}_L)$. Its diagonal entries $C_{d,d}(\boldsymbol{\psi}_L) = (-1)^{d-1} \prod_{s=1}^d \psi_L(s)$ are non-zero, so $C(\boldsymbol{\psi}_L)$ is invertible. Given the structural hazards from Step 1 (identified up to the scale $\mathbb{E}[\theta]$), the system $\boldsymbol{\mu} = C(\boldsymbol{\psi}_L)^{-1}\mathbf{g}_L$ identifies $\{\mathbb{E}[\theta^k]\}_{k=1}^{\bar{D}}$ up to the same scale.

Proof of Proposition 2. Consider the indicator product $\mathbb{I}\{\tilde{D} = d\}\mathbb{I}\{C = 0\}$ in the numerator. Since $C = 0$ implies $\tilde{D} = D$, this is equal to $\mathbb{I}\{D = d\}\mathbb{I}\{C = 0\}$. Further, with $D = d$ and $C = 0$ (completed duration is d and the spell is not censored), it must be that $D^c \geq d$. Hence, the numerator can be rewritten as

$$\mathbb{E}\left[\frac{\mathbb{I}\{L = \ell\}\mathbb{I}\{D = d\}\mathbb{I}\{D^c \geq d\}}{p_\ell(X)}\right] = \Pr(D^c \geq d) \cdot \mathbb{E}\left[\frac{\mathbb{I}\{L = \ell\}\mathbb{I}\{D = d\}}{p_\ell(X)}\right],$$

where the second equality follows from $D^c \perp (L, \nu, X)$. Similarly, for the denominator since $\tilde{D} = \min\{D, D^c\}$,

$$\mathbb{I}\{\tilde{D} \geq d\} = \mathbb{I}\{D \geq d\}\mathbb{I}\{D^c \geq d\},$$

so the denominator becomes

$$\mathbb{E}\left[\frac{\mathbb{I}\{L = \ell\}\mathbb{I}\{D \geq d\}\mathbb{I}\{D^c \geq d\}}{p_\ell(X)}\right] = \Pr(D^c \geq d) \cdot \mathbb{E}\left[\frac{\mathbb{I}\{L = \ell\}\mathbb{I}\{D \geq d\}}{p_\ell(X)}\right],$$

where again the second equality follows from D^c independence. Finally, dividing the numerator by the denominator, the common factor $\Pr(D^c \geq d)$ cancels, and the ratio of the remaining expectations equals $h_\ell(d)$ by Proposition 1.

APPENDIX B SEARCH MODEL CALIBRATION

I calibrate the model under standard values for model parameters. To maintain consistency with the econometric model, each period is assumed to be 12 weeks long. Corresponding to a 5 percent annual interest rate, the discount factor β is set equal to 0.985. I normalize the wage to 1 and set the replacement rate for unemployment benefits at 0.5. In addition, I assume individuals receive an annuity payment of 0.1 times their wages in each period, regardless of their employment status. This can be interpreted as the income of a secondary earner. Utility from consumption is given by the constant relative risk aversion (CRRA) utility function, $u(c) = c^{1-\sigma}/(1-\sigma)$ with $\sigma = 1.75$. I follow DellaVigna et al. (2017) and Marinescu and Skandalis (2021) in assuming that costs of job search are given by $c(s) = \xi s^{1+\rho}/(1+\rho)$. I set $\rho = 1$ and $\xi = 50$.²⁶ Table B1 summarizes the calibration parameters and Figure B1 displays the fit of the calibrated model.

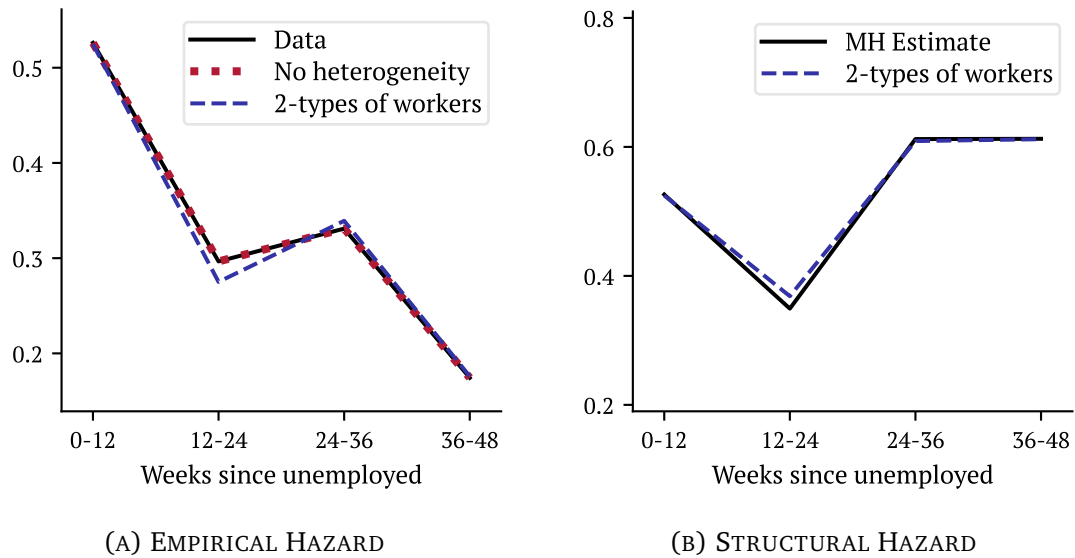
TABLE B1: CALIBRATION PARAMETERS FOR THE SEARCH MODEL

Parameter	Value
Length of each period	12 Weeks
Discount factor β	0.985
Relative risk aversion σ	1.75
Per period wages w	1
Annuity Payments	0.1
Unemployment benefits	0.5
Benefit exhaustion D_B	3
Search cost parameter ρ	1
Search cost parameter ξ	50
First period arrival rate $\delta(1)$	1

Note: The table presents the parameters used for calibrating the search model in Section VI.

²⁶Different parameters for the cost function do not change qualitative predictions of my exercise but do lead to changes in the scale of the search effort.

FIGURE B1: SEARCH MODEL CALIBRATION: FIT



Note: The figure displays the fit of the search model for the two calibration exercises described in the text. Panel A shows the exit rate in the data (solid black line) alongside the corresponding fitted values obtained from calibrating the search model without heterogeneity (dotted red line) and with two types of workers (dashed blue line). Panel B displays the estimated structural hazard from the Mixed Hazard (MH) model (solid black line) and the fitted structural hazard from calibrating the search model with two types of workers (dashed blue line).

APPENDIX C DATA

This section provides more details about data construction and sample selection. It also elaborates on the estimation of propensity scores and checks the balance of the weighted sample on additional measures. Supplementary data descriptives are also presented here.

C.1 Data Construction and Sample Selection

The Displaced Worker Supplement (DWS) was introduced in 1984, but the variable on the length of notice was not included in the first two samples. Furthermore, the definition of displaced workers has changed over time.²⁷ Before 1998, self-employed individuals or those who expected to be recalled to their lost job within six months were also included in the survey. However, the information on whether a worker expected to be recalled is only available for the years 1994 and 1996. In addition, the data on the length of time individuals took to find their next job is miscoded and largely missing for the year 1994. For these reasons, my analysis begins from 1996. Moreover, to maintain consistency across years, I exclude self-employed individuals or those who expected to be recalled from the 1996 sample.

The final sample only includes individuals with non-missing information on all the variables utilized in the analyses. Additionally, to capture individuals who had lost a job representing stable, non-temporary employment, the sample is limited to those who worked full-time for at least six months at their previous job and had health insurance provided through that employment. Finally, to minimize retrospective bias, I exclude individuals who report switching more than two jobs since losing their previous job. The sample selection procedure is summarized in Table C1.

The duration of unemployment for individuals who have secured a job by the time

²⁷The recall window was 5 years instead of 3 before 1994.

TABLE C1: SAMPLE SELECTION

Sample condition	Observations
DWS 1996-2020, 21-64 year old respondents	44707
No recall expectation	44537
Lost job was not self-employment	43471
Non-missing values for variables used	29443
Worked full-time at lost job	25293
Employed for at least 6 months at lost job	22991
Had health insurance at lost job	14825
Held less than 3 jobs since lost job	13784
Got a notice before job loss	5898
Got a notice of >1 month	4175

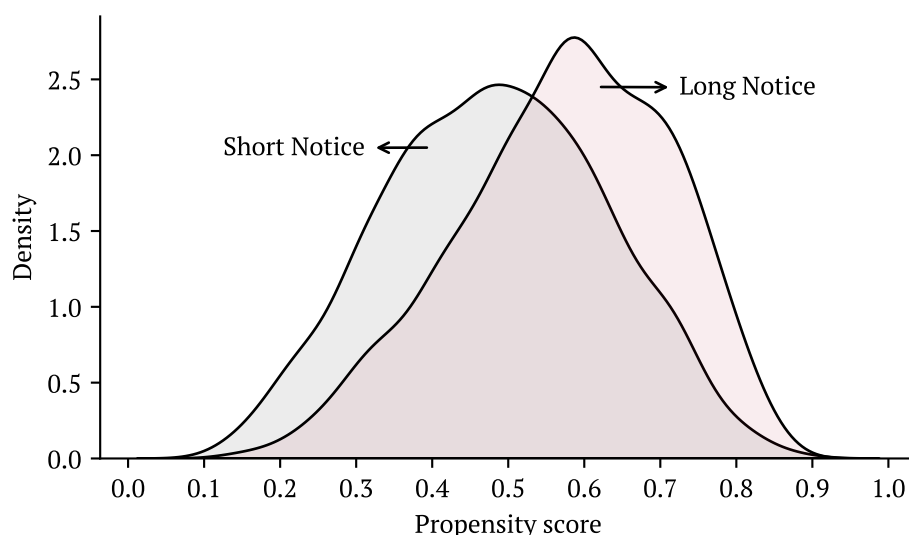
Note: The table shows the number of observations remaining at each step of sample selection.

TABLE C2: COMPARISON OF THE ANALYTICAL SAMPLE TO ALL INDIVIDUALS IN THE DIS-PLACED WORKER SUPPLEMENT (DWS) AND THE CURRENT POPULATION SURVEY (CPS)

	Sample (1)	DWS (2)	CPS (3)
Age	43.04	40.61	42.22
Female	0.45	0.44	0.52
Married	0.61	0.54	0.60
Black	0.10	0.11	0.10
High School	0.30	0.42	0.41
Some College	0.30	0.32	0.29
College Degree	0.40	0.26	0.30
Employed	0.80	0.67	0.75
Unemployed	0.18	0.21	0.04
NILF	0.02	0.12	0.21
Observations	4175	44707	964225

Note: All samples are restricted to individuals between the ages of 21 to 64 and pertain to years 1996-2020. Column (1) includes individuals from the DWS who worked full-time for at least six months and were provided health insurance at their lost job, did not expect to be recalled, and received a layoff notice of 1-2 months or greater than 2 months. Columns (2) and (3) include all individuals in the DWS and the monthly CPS, respectively.

FIGURE C1: ASSESSING OVERLAP OF PROPENSITY SCORE DISTRIBUTIONS



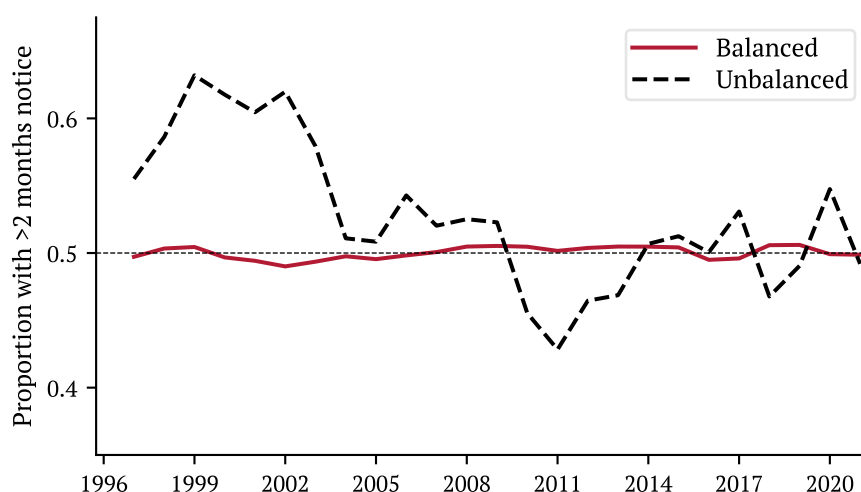
Note: The figure presents the density of estimated propensity scores for individuals with short and long notice separately.

of the survey is given by the *dwwksun* variable, which measures the number of weeks the person was unemployed between leaving or losing one job and starting another. For those who report not holding another job since their lost job, censored duration is obtained using the *durunemp* variable from the CPS. Since 2012, tenure at the lost job was top-coded at 24 years. To maintain consistency across samples, I also implement a top code of 24 years for all years prior to 2012. Earnings are reported in 1999 dollars. Table C2 presents the descriptive statistics for my analytical sample relative to all individuals aged 21 to 64 in the DWS and the CPS over the sample period.

C.2 Propensity Score Estimation

To ensure individuals with long and short notice are comparable, I reweight the sample using IPW. The weight for each individual is calculated as the inverse of the likelihood of receiving the reported notice length. To estimate the propensity scores, I use a logistic regression in which the odds of receiving a longer notice are modeled as a

FIGURE C2: LENGTH OF NOTICE OVER TIME

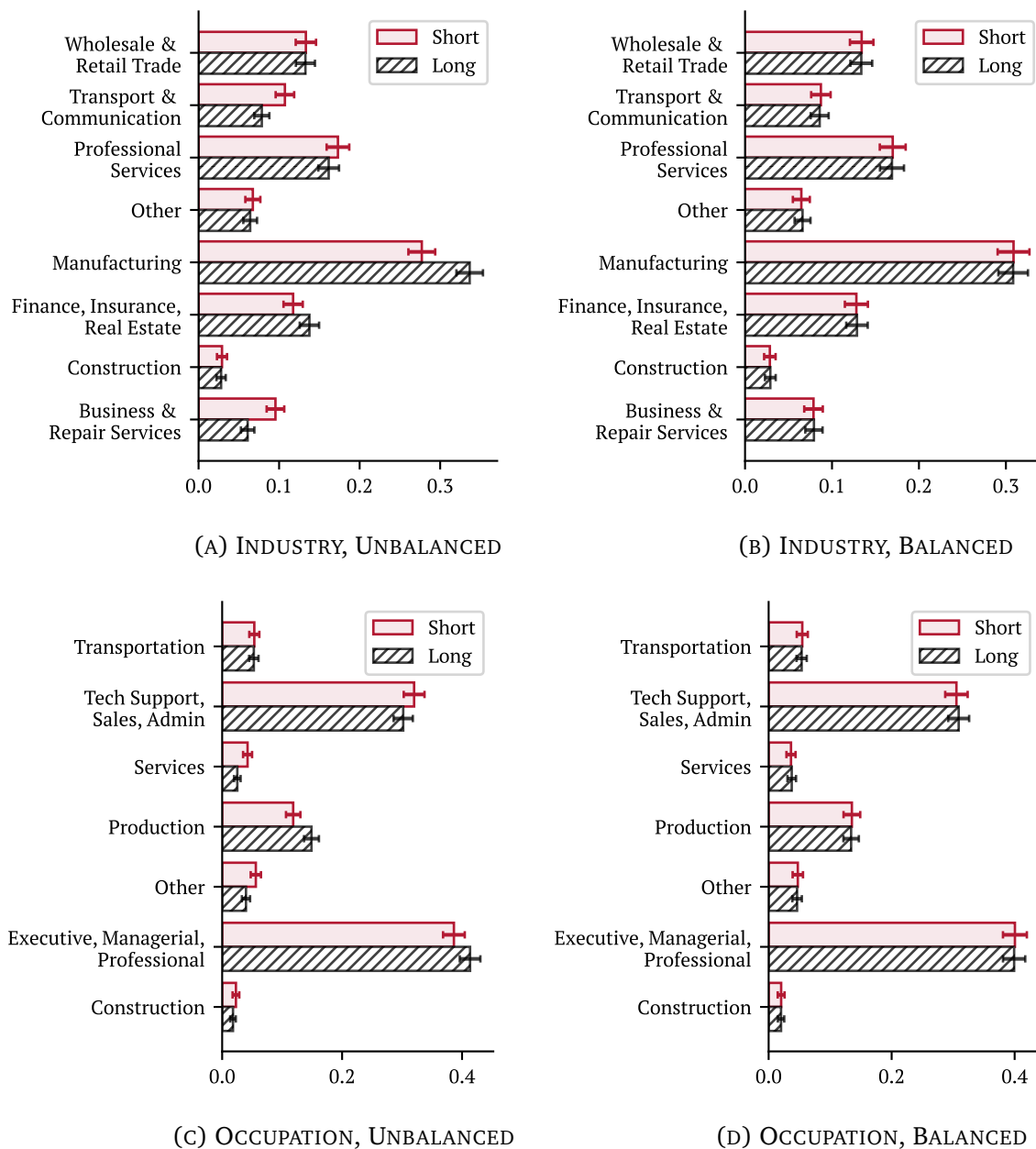


Note: The figure plots a 3-year moving average of the proportion of individuals who received a notice of more than 2 months amongst all individuals in the sample who were displaced in a given year.

function of several variables. These variables include age, gender, marital status, race (specifically, an indicator for Black), education level, reason for layoff, union membership, residence in a metropolitan area, tenure, and earnings at the lost job. Additionally, the model includes fixed effects for occupation and industry of the lost job, state of residence, and year of displacement. The density of estimated propensity scores for short and long-notice individuals is displayed in Figure C1. The figure indicates significant overlap between the two distributions, with all estimated propensity scores falling within the 0.1-0.9 range, rendering further data trimming unnecessary.

Table 1 in the main text provides evidence that the reweighting achieves balance across certain observable variables. Figure C2 demonstrates that reweighting leads to balance with respect to the year of displacement. In addition, Figure C3 presents occupation and industry distributions for short and long-notice workers in both the balanced and unbalanced samples. As expected, the weighted sample shows greater similarity in the industrial and occupational composition of short- and long-notice workers.

FIGURE C3: INDUSTRY AND OCCUPATION OF THE LOST JOB



Note: The figure presents the proportions of individuals whose displaced jobs were in specific industries (panels A and B) and occupations (panels C and D) among long-notice and short-notice workers in both the unbalanced and balanced samples. The error bars represent the 90% confidence intervals.

C.3 Additional Descriptives

This section provides additional descriptive statistics. Table C3 presents the relationship between longer notice and earnings at the subsequent job. The table shows that workers with longer notice periods tend to earn higher wages in their subsequent jobs. However, we cannot interpret this as a direct effect of longer notice, because extended periods of unemployment can negatively affect wages (Schmieder et al., 2016), and, as shown in this paper, longer notice leads to shorter unemployment spells. Table C4 describes the incidence of UI take-up in the sample, and Figure C4 describes the timing of benefit exhaustion amongst UI takers. Figure C5 presents the data with unemployment duration binned in 4 and 9-week intervals.

Finally, Figure C6 plots exit rates for short- and long-notice workers with and without adjusting for the compositional changes over the unemployment spell driven by observables. The adjustment is implemented via a Cox proportional hazards model estimated separately for each notice group, using the same set of controls as in the propensity score model in Section C.2. Note that this adjustment does not target differences in observables between the two notice groups. Instead, it accounts for how observables reshape the composition of survivors within each group over the unemployment spell. The Cox model is fit at the raw (unbinned) unemployment duration, and the resulting survival is aggregated to the 12-week intervals used in the main analysis. I recover an average adjusted hazard by evaluating the fitted survival function at each worker's covariates and averaging across the sample.

TABLE C3: EARNINGS AT THE SUBSEQUENT JOB

	Weekly Log Earnings			
	(1)	(2)	(3)	(4)
> 2 month notice	0.091** (0.037)	0.066** (0.032)	0.093** (0.038)	0.063* (0.032)
Controls	No	Yes	No	Yes
Weights	No	No	Yes	Yes
Observations	2657	2657	2657	2657

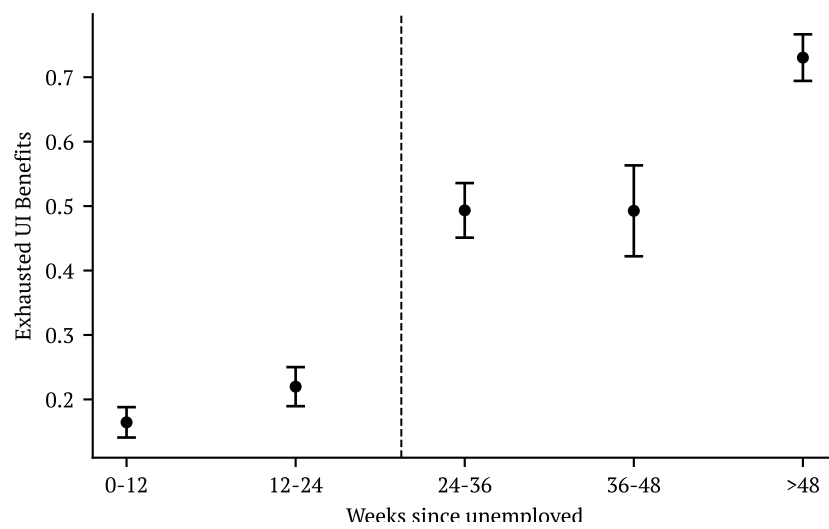
Note: The table shows results from linear regressions of log weekly wages at the subsequent job on an indicator for receiving a notice of more than 2 months. The sample used is similar to the main analytical sample, but it excludes individuals who had not yet found employment at the time of the survey, had multiple jobs between their previous and current jobs, or had incomplete earnings information for other reasons. Robust standard errors are reported in parentheses.

TABLE C4: UNEMPLOYMENT INSURANCE TAKE-UP

Duration	Observations	Received UI Benefits
0 weeks	820	0.06
0-4 weeks	959	0.30
4-8 weeks	457	0.61
8-12 weeks	363	0.70
> 12 weeks	1562	0.82

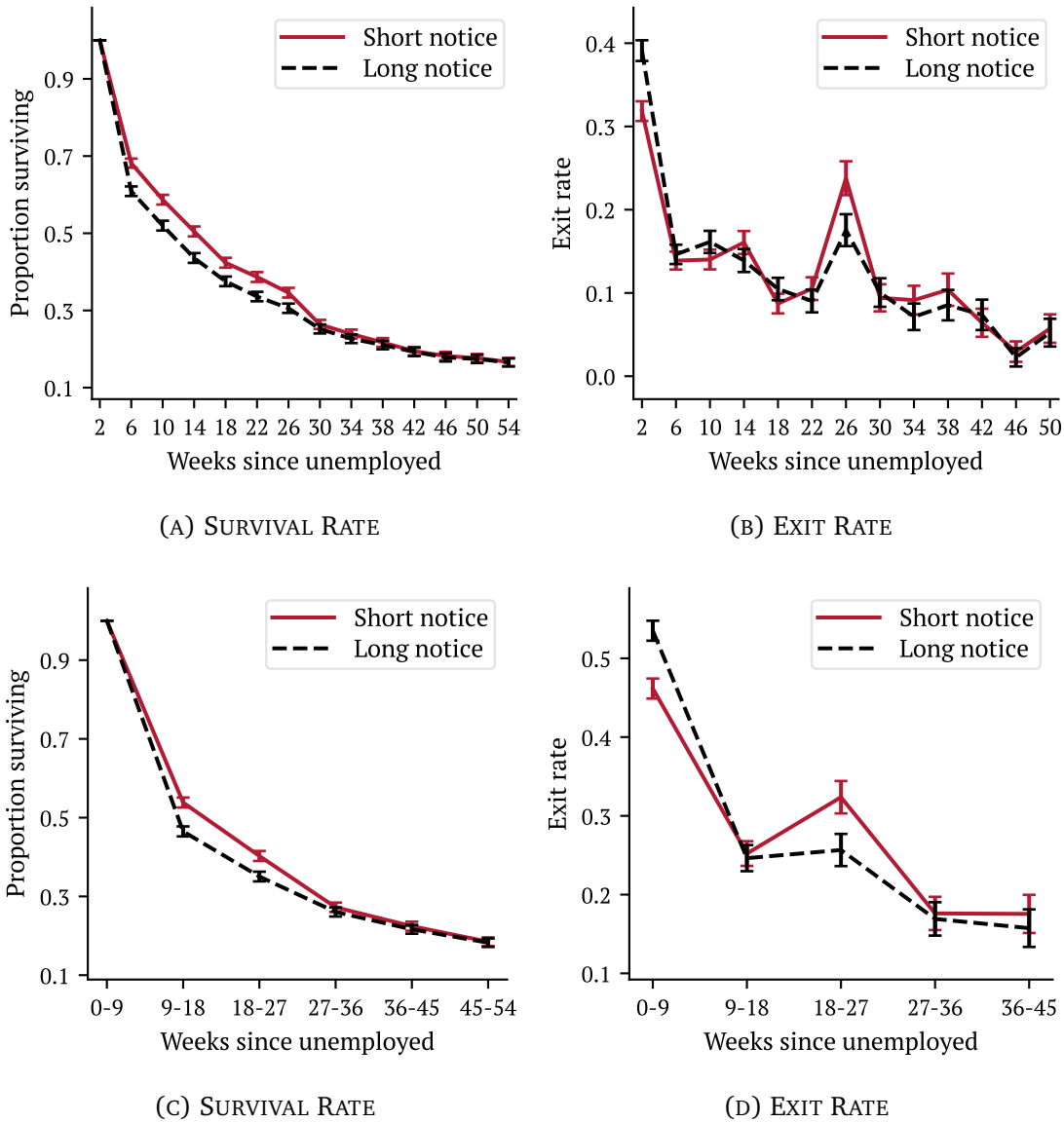
Notes: This table reports the percentage of individuals in the baseline sample who reported receiving UI benefits by the duration of unemployment.

FIGURE C4: TIMING OF BENEFIT EXHAUSTION



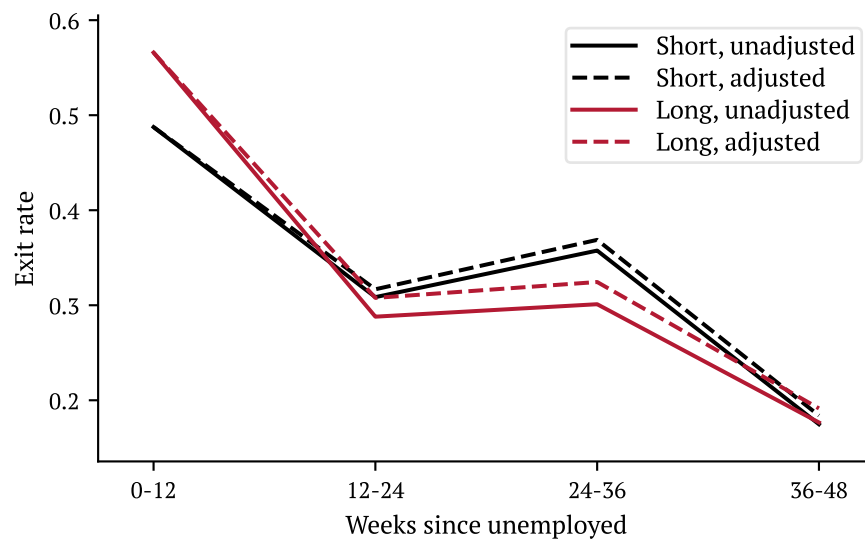
Note: The figure presents the proportion of individuals who report having exhausted their UI benefits by the duration of unemployment. The sample is restricted to individuals in the main analytical sample who reported receiving UI benefits, and duration is binned in 12-week intervals.

FIGURE C5: SURVIVAL AND EXIT RATES WITH ALTERNATIVE BINS



Note: Unemployment duration is binned in 4-week intervals for panels A and B, while it is binned in 9-week intervals for panels C and D. Panel A and C present the proportion of individuals who are unemployed at the beginning of each interval. Panel B and D present the proportion of individuals exiting unemployment in each interval amongst those who were still unemployed at the beginning of the interval. Error bars represent 90% confidence intervals.

FIGURE C6: EFFECT OF OBSERVABLE COMPOSITION CHANGES ON EXIT RATES



Note: The figure plots exit rates by 12-week interval for short- and long-notice workers, both without adjustment (solid) and after adjusting for the compositional changes over the unemployment spell driven by observables (dashed). The adjustment is obtained from a Cox proportional hazards model estimated separately for each notice group, using the same set of controls as in the propensity score model in Section C.2.

APPENDIX D ROBUSTNESS CHECKS

In this subsection, I present a series of robustness checks.

D.1 Non-Parametric Estimates

As shown in the paper, the specified MH model is non-parametrically identified. Figure D1 presents the non-parametric estimate for the structural hazard alongside the baseline estimate, which assumes a log-logistic functional form for the hazard. This figure shows that the non-parametric point estimates are roughly equivalent to the baseline estimates. However, the standard errors are larger, particularly the one for the last data point, which blows up dramatically.

D.2 Unweighted Sample

Figure D2 displays the data and estimates using the unweighted sample. Panel A shows that the pattern of exit rates and how they vary by notice length is similar to that observed in the weighted sample in Figure 2. Similarly, the estimates based on the unweighted sample, presented in panel B, closely resemble the baseline estimates shown in Figure 3. This suggests that observable characteristics have a limited role in explaining the raw differences in exit rates between the two notice groups.

D.3 Alternative Notice Categories

As noted in Section II, the question on notice length in the DWS is categorical and includes categories of no notice, <1 month, 1-2 months, and >2 months. So far, I have only focused on the latter two categories, referred to as short and long notice. I now test the robustness of my findings with respect to this restriction. First, I include individuals who received no notice as a third group in the original estimation sample. Then, I

repeat the process, adding individuals with <1 month's notice as the third group.²⁸ I use the same covariates as before to fit the propensity score models using multinomial logit and weight the data accordingly. Observations where the propensity scores fall outside the 0.1 to 0.9 range are excluded. I then re-estimate the Mixed Hazard model with three notice categories for both the augmented samples.

The estimates from the above exercise are presented in Figure D3. Panel A corresponds to the sample where the third category is no notice, while panel B corresponds to the sample where the third category is <1 month notice. After excluding extreme propensity score values, 7,122 of 12,061 observations remain for the data used in panel A, and 5,667 of 5,898 observations remain for the data used in panel B. In both cases, the estimates of structural duration dependence are qualitatively similar to the baseline results—the structural hazard consistently exceeds the exit rate from the data, initially decreasing, then increasing up to the third interval, and remaining constant thereafter. However, in panel B, the estimated structural hazard increases less leading up to benefit exhaustion than in the baseline estimate. This could reflect differences in patterns of heterogeneity or duration dependence across the samples. More importantly, because we have more than two notice categories, the model is over-identified. With $3 \times 4 = 12$ moment conditions and 8 parameters, there are 4 degrees of freedom. The Sargan-Hansen J-statistic for testing overidentifying restrictions is reported in both panels, along with the respective p -values.²⁹ In both cases, the large p -values indicate that there is no strong evidence against the null hypothesis.

²⁸I do not include all four categories simultaneously for two reasons. Firstly, the overlap in propensity scores is already suboptimal with three categories and deteriorates further with four. Secondly, the exit rate in the first period for the <1 month group is lower than that for the *no notice* group, possibly due to selection, since separations with no notice may also include voluntary quits. However, the first-period exit rates for both are lower than those for the 1-2-month and > 2 -month notice groups.

²⁹The estimated model in the main analysis is also over-identified due to the imposition of a log-logistic functional form on the structural hazard, resulting in one free parameter. However, here we are testing a more extensive set of overidentifying restrictions, with the caveat that the power of this test is lower.

D.4 Binning Unemployment Duration

Finally, a limitation of the application in the paper is the need to bin unemployment durations into 12-week intervals due to the small sample size of the DWS and the potentially noisy measurement of duration caused by retrospective bias. While Figure C5 shows that the patterns of how exit rates vary with duration and notice length are similar when binning data into 4- or 9-week intervals, I now present estimates from the MH model using unemployment duration data binned into 9-week intervals in Figure D4. Once again, we uncover the same pattern in estimates of structural duration dependence, with a decline from the first to the second interval, followed by a rise in the third interval corresponding to 18-27 weeks, which aligns with UI exhaustion at 26 weeks. I also present non-parametric estimates, which closely follow the log-logistic estimates.

I conduct another exercise to evaluate the effect of interval sizes directly on estimates from the MH model. Specifically, I take exit rates implied by the model for a given data-generating process (DGP) and compute the exit rates that would be observed if unemployment durations were binned into different-sized intervals. I then compare how the estimates change with the size of the bins.

The data generating process (DGP) for this exercise is defined as follows: There are two groups of notice, each with a probability of 0.5, and the first-period structural hazards are set at 0.1 and 0.2 for the two groups, respectively. For periods after the first ($d > 1$), the structural hazards, which are common to both groups, are generated in four different cases using the Weibull function bkd^{k-1} to ensure our conclusions are not an artifact of a specific DGP. These four cases are: (1) increasing ($b = 0.2, k = 1.2$), (2) decreasing ($b = 0.2, k = 0.75$), (3) constant ($b = 0.15, k = 1$), and (4) non-monotonic, where the Weibull function is increasing as in (1) until $d = 7$ and then follows (2) multiplied by 1.75. The unobserved type ν is a mixture of three Beta distributions: $\nu_1 \sim \text{Beta}(0.1, 0.1)$ with probability 0.5, $\nu_2 \sim \text{Beta}(0.3, 0.5)$ with probability 0.1, and

$\nu_3 \sim \text{Beta}(0.25, 0.5)$ with probability 0.4. I generate the exit rates for 12 time periods, bin them into intervals of four different sizes—1, 2, 3, and 4—and estimate the model in each case.

The dashed black line in Figure D5 represents the profile of exit rates corresponding to different bin sizes used in the estimation for the four DGPs from panels A-D. The solid dark-grey line shows the estimated structural hazard in each case. Comparing the estimates across different bin sizes in each panel, one can see that increasing the bin size leads to potentially missing the full extent of changes in the structural hazard over time. Specifically, when the structural hazard is increasing, the binned estimate increases by less (compare column 4 to column 1 in panel A). Similarly, when the structural hazard is decreasing, the binned estimate decreases by less (panel B). This is a direct consequence of the binned exit rates not fully capturing the extent of changes in the exit rates over time, making it difficult for the estimates using those exit rates to capture the full dynamics.

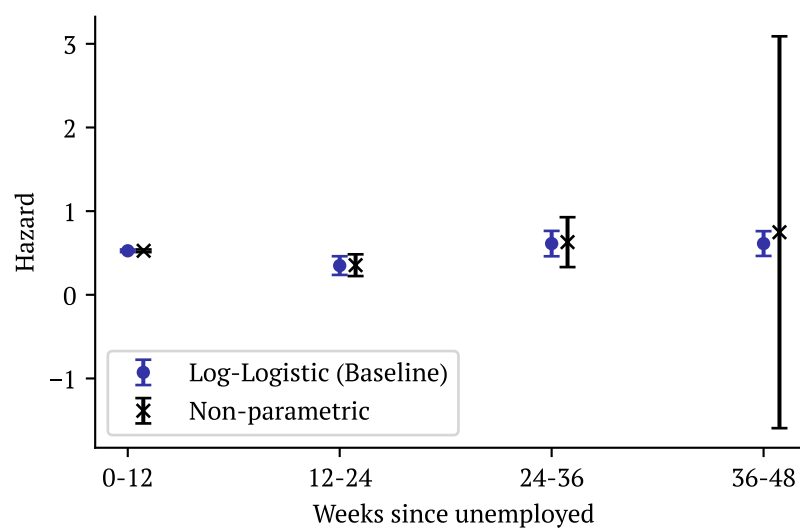
However, note that regardless of bin size, the estimates are able to uncover the broader trends in the structural hazard. This is because the estimates correspond to the cumulative structural hazard within a given interval for the average type, represented by the dotted red line on the plots. Specifically, the cumulative structural hazard is defined as the probability of an individual exiting unemployment in a given interval, provided that the individual hasn't exited by the start of the interval.³⁰ So, while the estimates may miss some nuances within the intervals, the extent of what is missed is limited to what would be missed if we were analyzing the true structural exit probabilities in broader intervals. This is also evident from the implied average type for each estimate presented in Figure D6. From this figure, it can be seen that the average type in each panel falls similarly over time for different bin sizes, implying that we are able

³⁰For instance, for a bin size of 2, the cumulative structural hazard corresponding to the second interval for the average worker is given by: $\Pr(3 \leq D \leq 4 | D \geq 2, L, \nu = \mu) = \psi(3)\mu + [1 - \psi(3)\mu]\psi(4)\mu$.

to capture the contribution of underlying heterogeneity in the exit rates very closely, regardless of the bin size.

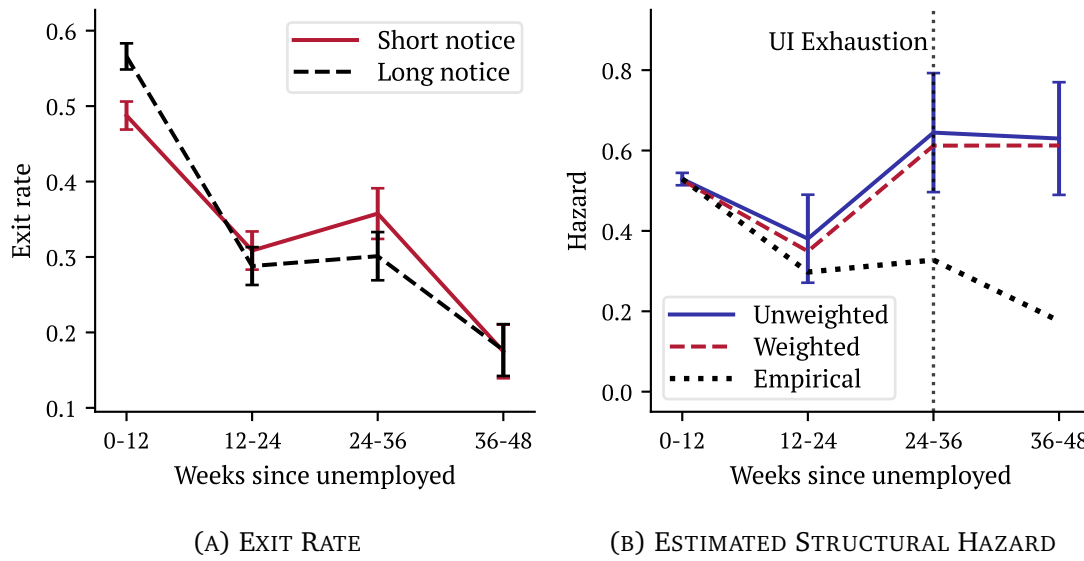
Overall, this shows that the assumptions of the model do not interact with bin sizes in a way that leads to systematically biased estimates, ensuring that the estimates reported in this paper correspond to meaningful underlying patterns.

FIGURE D1: NON-PARAMETRIC ESTIMATES



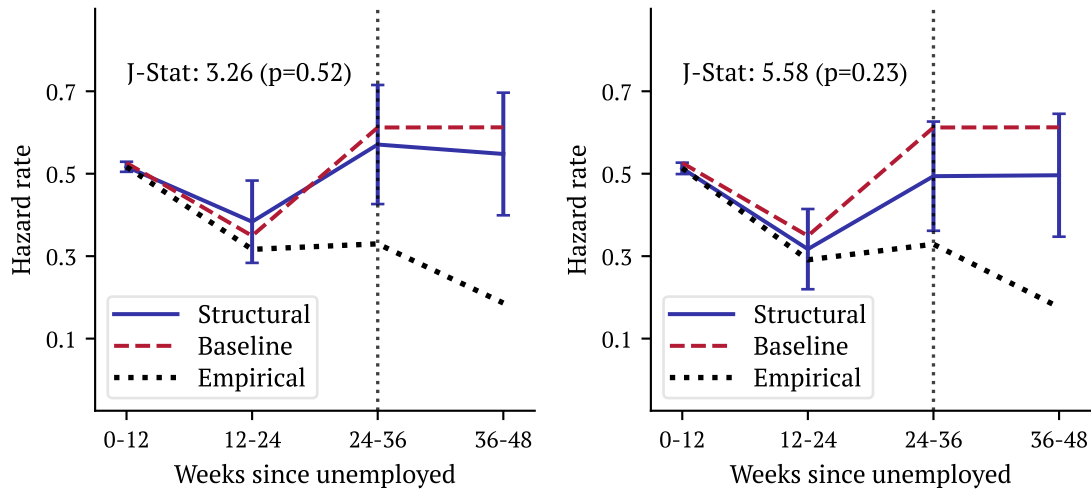
Note: The figure compares non-parametric estimates of the structural hazard from the Mixed Hazard model with baseline estimates that assume a log-logistic functional form for the structural hazard. Error bars represent 90% confidence intervals.

FIGURE D2: DATA AND ESTIMATES USING THE UNWEIGHTED SAMPLE



Note: The figure presents data and estimates for the unweighted analytical sample. Panel A displays the exit rate from the data separately for long and short-notice workers. In panel B, the solid blue line shows the estimated structural hazard from the Mixed Hazard model using unweighted moments, the dashed red line shows the estimate using weighted moments, and the dotted black line represents the average exit rate for short and long-notice workers in the data.

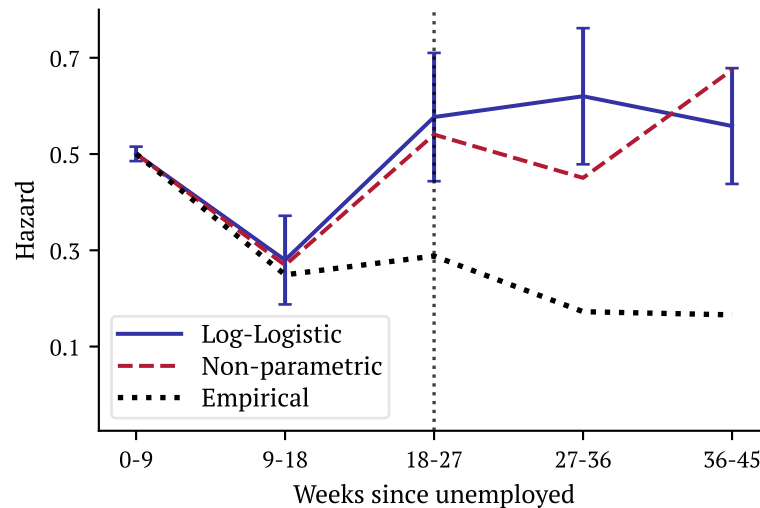
FIGURE D3: ESTIMATES WITH ALTERNATIVE NOTICE CATEGORIES



(A) NO NOTICE, 1-2 MONTHS, >2 MONTHS (B) <1 MONTH, 1-2 MONTHS, >2 MONTHS

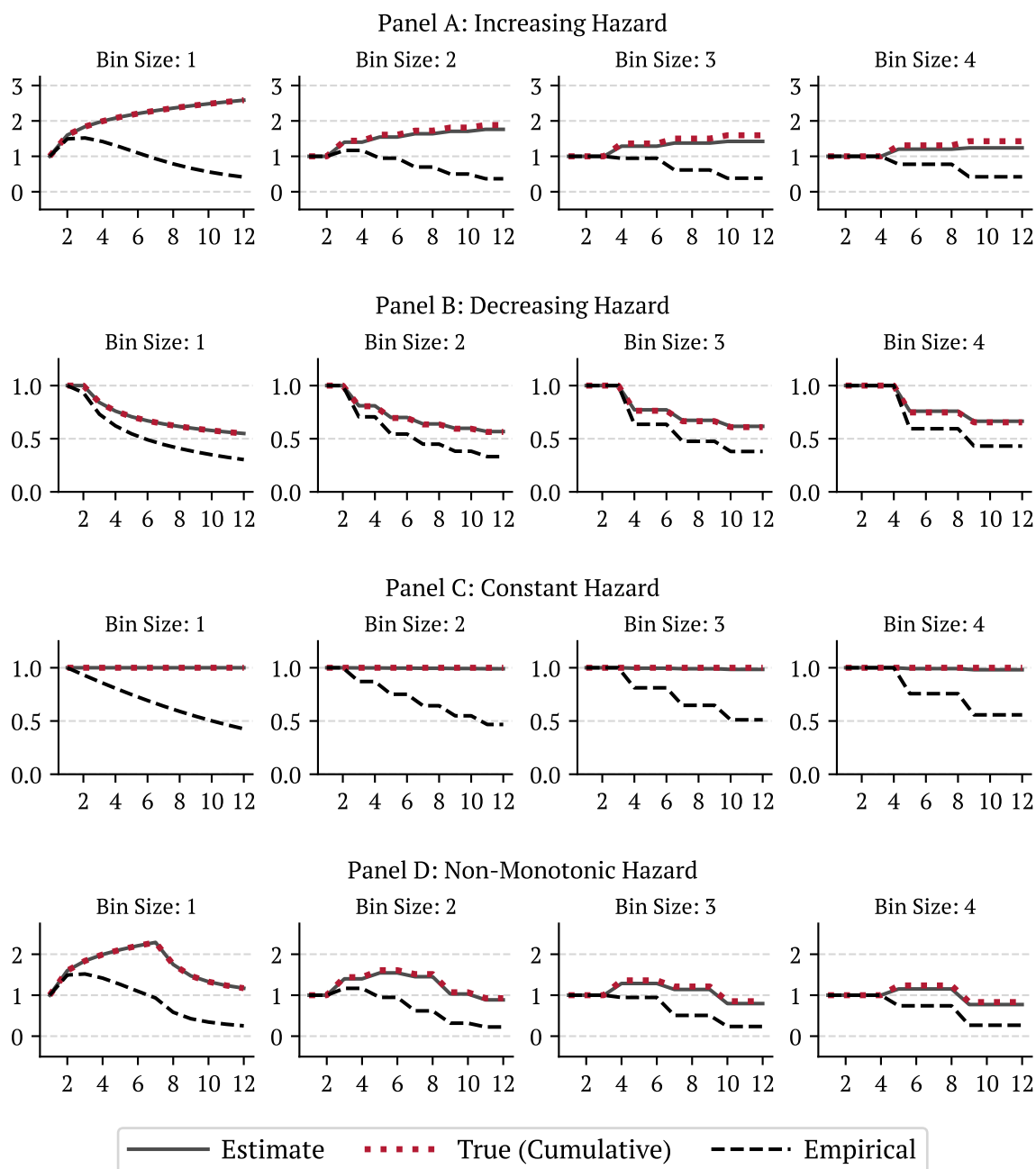
Note: Panel A presents the estimated hazard for the sample consisting of individuals who received no notice, along with those who received notice periods of 1-2 months and >2 months, as in the main analytical sample. Panel B estimates are for individuals who received notice of <1 month, in addition to the groups from the main analytical sample. J-Stat refers to the Sargan-Hansen statistic for overidentifying restrictions.

FIGURE D4: ESTIMATES WITH UNEMPLOYMENT DURATION BINNED IN 9-WEEK INTERVALS



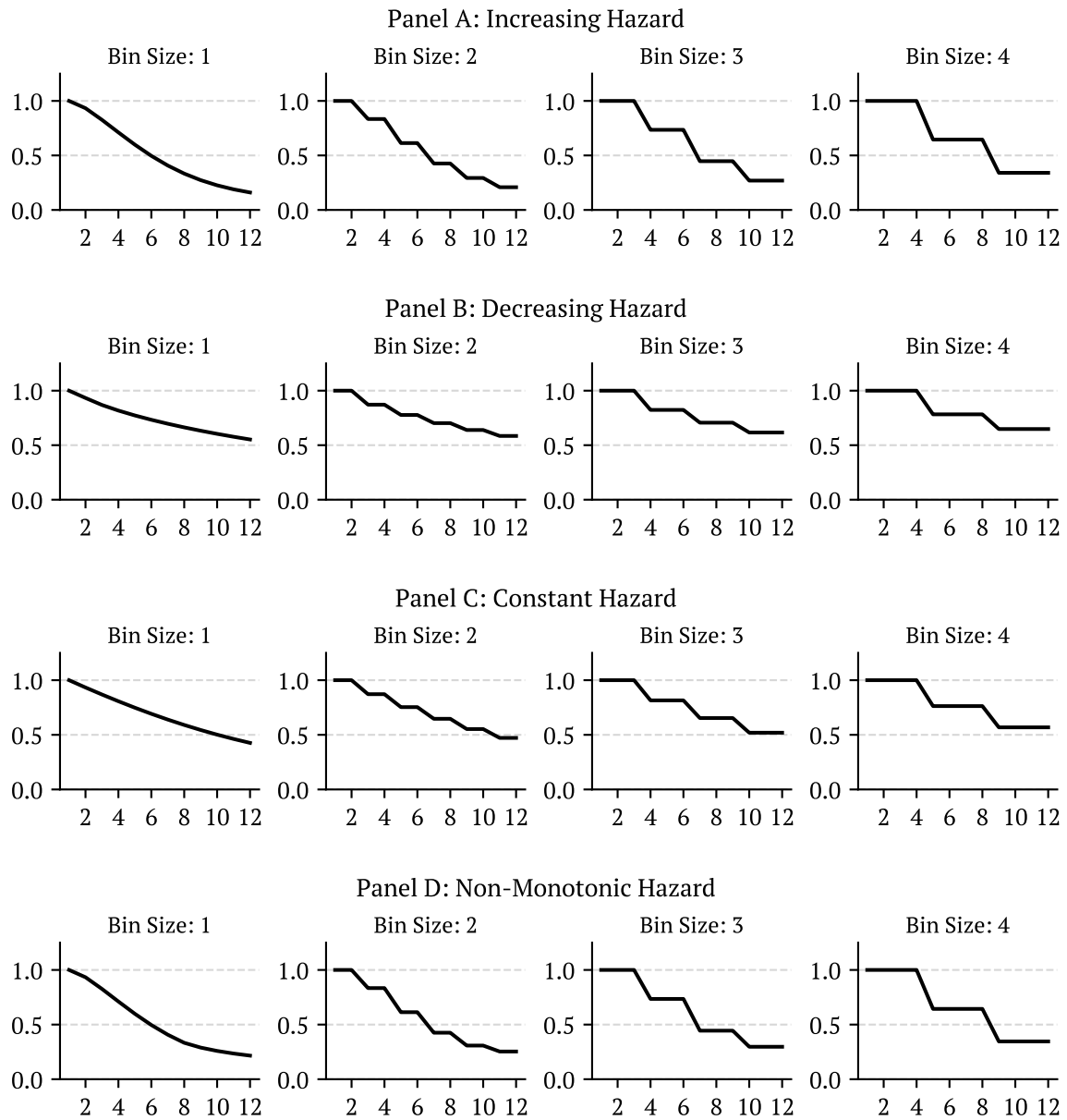
Note: The figure presents estimates from the Mixed Hazard model using data with unemployment duration binned in 9-week intervals. The solid blue line represents estimates for the structural hazard assuming the Log-Logistic functional form. The dashed red line represents non-parametric estimates, while the dotted black line represents the exit rate from the data.

FIGURE D5: ESTIMATION ON BINNED SIMULATED DATA: HAZARD



Note: The figure presents results from a robustness exercise evaluating the impact of binning data into intervals. I calculate the moments used for estimation from the Mixed Hazard model with 12 durations. These moments are then binned into intervals of lengths ranging from 1 to 4, and the model is estimated using the binned moments. Along with the estimates of the structural hazard (dark grey solid line), I present two additional quantities binned similarly for comparison. The dotted red line represents the true structural cumulative hazard of exiting in the specific interval, while the black dashed line represents the average empirical hazard.

FIGURE D6: ESTIMATION ON BINNED SIMULATED DATA: AVERAGE TYPE



Note: The figure presents results from a robustness exercise evaluating the impact of binning data into intervals. I calculate the moments used for estimation from the Mixed Hazard model with 12 durations. These moments are then binned into intervals of lengths ranging from 1 to 4, and the model is estimated using the binned moments. The plots represent the average type at each duration, as implied by the estimated structural hazard.

APPENDIX E SENSITIVITY TO THE IDENTIFYING ASSUMPTIONS

In this section, I examine the implications of relaxing Assumption 2 (conditional independence) and Assumption 3 (stationarity). I consider two deviations from the baseline framework: a scale difference κ in the type distribution across notice groups, and a scale difference γ in the structural hazards beyond the first period. I introduce these deviations and the accompanying notation, and show that they are observationally equivalent, entering only through a single composite parameter $\eta = \gamma\kappa$. I then use the support and moment restrictions implied by a valid type distribution to derive bounds on η , and report the estimates these bounds imply.

E.1 Two Deviations from Baseline

To simplify the exposition here, I write an individual worker's exit probability as $h(d \mid \theta_L, L) = \psi_L(d)\theta_L$, where the type θ_L captures both observable and unobservable characteristics. Throughout, let ℓ denote the short-notice and ℓ' denote the long-notice.

First, I allow the type distribution to differ across notice groups by a scale factor, $\theta_{\ell'} \stackrel{d}{=} \kappa\theta_{\ell}$. Writing $\mu_{L,k} \equiv \mathbb{E}[\theta_L^k]$ for the k th moment of θ_L , the scale shift implies $\mu_{\ell',k} = \kappa^k \mu_{\ell,k}$. When $\kappa > 1$, the long-notice group is drawn from a proportionally higher type distribution, corresponding to positive selection into long notice. This rescaling changes the level of the type distribution but not its shape. The standardized moments, which divide the k th moment by the k th power of the mean and are therefore invariant to scale, are common across groups,

$$\lambda_k \equiv \frac{\mu_{\ell,k}}{\mu_{\ell,1}^k} = \frac{\mu_{\ell',k}}{\mu_{\ell',1}^k},$$

since the factors of κ cancel. For instance, the mean scales by κ and the variance by κ^2 , so the long-notice group's types are larger and more dispersed in absolute terms, but their coefficient of variation, which captures dispersion relative to the mean, is

unchanged.³¹

Second, I allow the structural hazards to differ across groups beyond the first period, $\psi_{\ell'}(d) = \gamma\psi_{\ell}(d)$ for $d > 1$. When $\gamma > 1$, longer notice raises the structural hazard later in the spell, as would arise if time spent searching makes workers more effective through learning while searching (Burdett and Vishwanath, 1988; Gonzalez and Shi, 2010). On the other hand, when $\gamma < 1$, the exit probability falls with time spent searching, as in the stock-flow model of job search (Coles and Smith, 1998) or if workers become discouraged and lower their search effort over time.

E.2 Observational Equivalence

I now show that these two deviations are observationally equivalent: beyond the first period, κ and γ enter the exit probabilities only through their product $\eta = \gamma\kappa$, so the duration data cannot identify them separately. The recovered structural hazards and moments of the type distribution then depend only on η , and the two interpretations differ only in the implied effect of notice in the first period.

Without loss of generality, normalize the mean of the short-notice group to $\mu_{\ell,1} = 1$, so that $\lambda_k = \mu_{\ell,k} = \mu_{\ell',k}/\kappa^k$. Now consider the exit probabilities $g(d|L) = \Pr(D = d|L)$ for the first two periods. For $d = 1$,

$$g(1|\ell) = \psi_{\ell}(1), \quad g(1|\ell') = \kappa\psi_{\ell'}(1),$$

so $\psi_{\ell}(1) = g(1|\ell)$ and $\psi_{\ell'}(1) = g(1|\ell')/\kappa$. Substituting these first-period hazards into

³¹These models identify the type distribution only up to an overall scale, so the between-group difference is naturally expressed as the ratio $\kappa = \mu_{\ell',1}/\mu_{\ell,1}$, which is invariant to the normalization, rather than as an additive shift, whose magnitude would depend on the units chosen for θ .

the second-period equations gives,

$$g(2|\ell) = \psi_\ell(2)[1 - g(1|\ell)\lambda_2],$$

$$g(2|\ell') = \eta\psi_\ell(2)[1 - g(1|\ell')\lambda_2],$$

The parameters κ and γ enter only through the product η because both scale the same object, $\psi_L(d)\theta_L$: κ scales the type component θ_L , while γ scales the structural hazard component $\psi_L(d)$. Consequently, the data identify only their combined effect, $\eta = \gamma\kappa$, rather than κ and γ separately. This is true at all durations beyond the first period. In the first period γ does not operate, and the long-notice hazard is the product $g(1|\ell') = \kappa\psi_{\ell'}(1)$, identified directly from the data; the later periods depend only on this identified product, not on how it splits into κ and $\psi_{\ell'}(1)$, so the one quantity the data leave undetermined is confined to the first period. That split is what distinguishes the two readings: under pure non-stationarity ($\kappa = 1$, $\eta = \gamma$) the entire first-period hazard is the notice effect, $\psi_{\ell'}(1) = g(1|\ell')$, whereas under positive selection ($\kappa > 1$) part of it is composition and the notice effect is smaller, $\psi_{\ell'}(1) = g(1|\ell')/\kappa$.

E.3 Direction of the Bias

Before turning to the direction of the bias, note that the model is identified with fixed η . The two second-period equations above contain three unknowns, $\psi_\ell(2)$, λ_2 , and η . Once η is fixed, they form a system of two equations in two unknowns, which identifies $\psi_\ell(2)$ and λ_2 . The same recursive structure applies in every subsequent period. In period d , $g(d|\ell)$ and $g(d|\ell')$ depend on η , on the structural hazards and moments recovered from earlier periods, and on two new unknowns: the period- d structural hazard $\psi_\ell(d)$ and the moment λ_d . Thus, conditional on η , each period is just-identified, and the recursion recovers all structural hazards and moments of the type distribution

as functions of η .³²

In particular, dividing the two second-period equations to eliminate $\psi_\ell(2)$ gives λ_2 as a function of η ,

$$\lambda_2 = \frac{g(2|\ell') - \eta g(2|\ell)}{g(1|\ell)g(2|\ell') - \eta g(1|\ell')g(2|\ell)}.$$

The structural hazard then follows from the short-notice equation,

$$\psi_\ell(2) = \frac{g(2|\ell)}{1 - g(1|\ell)\lambda_2},$$

which increases with λ_2 . Since $\lambda_2 = 1 + \text{Var}(\theta_\ell)$ measures the dispersion of types, a larger recovered heterogeneity implies a larger recovered structural hazard: with more heterogeneity, more of the observed decline in exit rates is attributed to dynamic selection, so the underlying hazard must be higher to match the data.

To sign the bias, it is therefore enough to see how λ_2 varies with η . Differentiating the expression for λ_2 ,

$$\frac{d\lambda_2}{d\eta} = \frac{g(2|\ell)g(2|\ell')(g(1|\ell') - g(1|\ell))}{(g(1|\ell)g(2|\ell') - \eta g(1|\ell')g(2|\ell))^2}.$$

Given that $g(1|\ell') > g(1|\ell)$, we have $d\lambda_2/d\eta > 0$: the recovered heterogeneity is increasing in η . Combined with the previous step, the recovered structural hazard increases in η as well, so the baseline ($\eta = 1$) understates both heterogeneity and the structural hazards when the truth has $\eta > 1$, and overstates them when $\eta < 1$.

To see the intuition, write the long-to-short ratio of second-period equations as η times the depletion factor $[1 - g(1|\ell')\lambda_2]/[1 - g(1|\ell)\lambda_2]$. Because the long group exits faster in the first period, this factor is below one and falls as λ_2 rises: a more dispersed type distribution depletes the faster-exiting group of high types more severely, pulling its second-period exit down. The structural advantage η pushes the same ratio up.

³²A formal identification proof is omitted because it follows the recursive argument described in the text and adds little additional insight here.

Since the ratio is observed, the two must move together: a larger η has to be offset by a larger λ_2 to reproduce it. Imposing $\eta = 1$ therefore selects the smallest λ_2 consistent with the data, so when the truth has $\eta > 1$ the baseline understates heterogeneity, and with it the structural hazards.

One thing to note is that the bias depends only on η , not on how it splits between the two channels. When the channels push the same way (κ and γ both above, or both below, one), η lies on that side of one. When they oppose ($\kappa > 1$ with $\gamma < 1$, or $\kappa < 1$ with $\gamma > 1$), η can fall on either side, according to which channel dominates. In particular, $\eta > 1$ does not require positive selection: negative selection ($\kappa < 1$) paired with a sufficiently larger later hazard ($\gamma > 1$) also yields $\eta > 1$. The data therefore sign the bias through η but do not identify the selection channel, so $\eta > 1$ is consistent with positive selection, with pure non-stationarity, and with negative selection offset by a larger structural increase. The direction of the bias is the same in all of these, since it depends only on η .

E.4 Bounding the Composite Parameter

Lower bound. As argued above, given η , the recursion delivers the moments of the type distribution, so η is admissible only if these are the moments of a genuine distribution of positive types. In particular, we require the following Hankel matrix, and its shift to be positive semidefinite. Because the application has four periods, the data identify the first four moments of the type distribution, and admissibility reduces to:

$$H = \begin{pmatrix} 1 & 1 & \lambda_2 \\ 1 & \lambda_2 & \lambda_3 \\ \lambda_2 & \lambda_3 & \lambda_4 \end{pmatrix} \succeq 0 \quad \text{and} \quad \begin{pmatrix} 1 & \lambda_2 \\ \lambda_2 & \lambda_3 \end{pmatrix} \succeq 0.$$

The binding conditions implied by above are:

$$\lambda_2 \geq 1, \quad \lambda_3 \geq \lambda_2^2, \quad \lambda_4 \geq \underline{\lambda}_4 \equiv \frac{\lambda_3^2 - 2\lambda_2\lambda_3 + \lambda_2^3}{\lambda_2 - 1},$$

where the last condition follows from $\det H \geq 0$.

We have shown that $\lambda_2(\eta)$ is increasing in η , so the variance condition $\lambda_2(\eta) \geq 1$ delivers a lower bound on η : the value at which $\lambda_2(\eta) = 1$. This bound is not informative, since $\lambda_2 = 1$ is the degenerate case of no heterogeneity. The higher-moment conditions tighten it. Their closed forms in η grow unwieldy beyond λ_2 , so I impose them numerically. For a grid of values of η , the recursion recovers the implied moments $\lambda_2(\eta)$, $\lambda_3(\eta)$, and $\lambda_4(\eta)$, and I check the Hankel conditions $\lambda_2 \geq 1$, $\lambda_3 \geq \lambda_2^2$, and $\lambda_4 \geq \underline{\lambda}_4$ at each. The lower bound $\underline{\eta}$ is the smallest η at which all three hold.

Figure E1 plots the three conditions against η , one per panel: $\lambda_2(\eta)$ against one, $\lambda_3(\eta)$ against $\lambda_2(\eta)^2$, and $\lambda_4(\eta)$ against the floor $\underline{\lambda}_4(\eta)$. As η rises the conditions are met in turn: the variance condition from $\eta \approx 0.93$, the third-moment condition from $\eta \approx 1.07$, and the fourth-moment condition from $\eta \approx 1.12$. Each curve crosses its threshold once, so each condition holds on an interval of η and admissibility, their intersection, is itself an interval; it requires η above the largest crossing. The fourth-moment condition binds last, so it sets the lower bound, $\underline{\eta} \approx 1.12$.

Upper bound. The upper bound comes from the type having bounded support, $\theta_\ell \in (0, \bar{\theta}]$. No individual hazard can exceed one; for the long group in the second period this requires $\eta\psi_\ell(2)\bar{\theta} \leq 1$. At the same time, any distribution satisfies $\bar{\theta} \geq \mu_{\ell,2}/\mu_{\ell,1} = \lambda_2$.³³ Combining gives $\lambda_2 \leq 1/(\eta\psi_\ell(2))$. Using the long-group second-period equation $g(2|\ell') = \eta\psi_\ell(2)[1 - g(1|\ell')\lambda_2]$ to eliminate $\eta\psi_\ell(2)$, this becomes $\lambda_2 \leq [1 - g(1|\ell')\lambda_2]/g(2|\ell')$, which rearranges to $\lambda_2 \leq \bar{\lambda}_2 \equiv 1/(g(1|\ell') + g(2|\ell'))$. Inverting the $\lambda_2(\eta)$ expression and plugging in $\bar{\lambda}_2$, we get $\bar{\eta} = 1.29$.

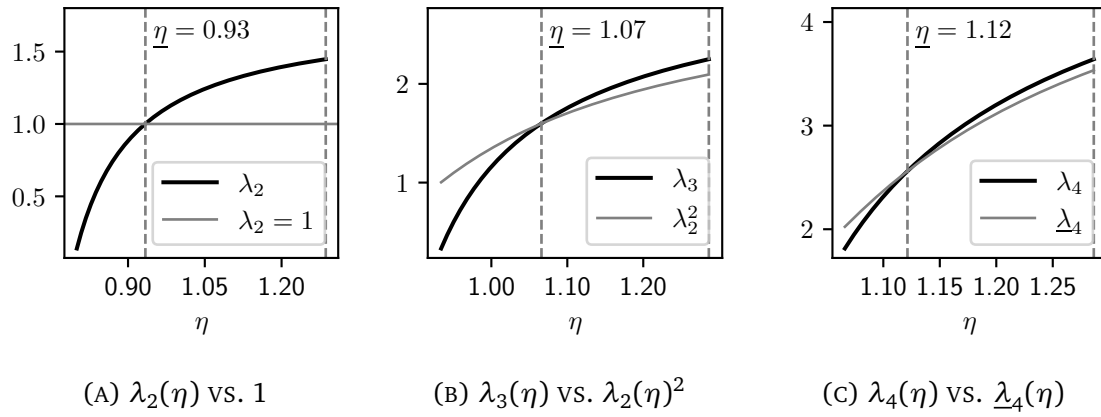
The final identified set is: $1.12 \leq \eta \leq 1.29$.

³³Since $\theta_\ell(\bar{\theta} - \theta_\ell) \geq 0$, taking expectations gives $\bar{\theta}\mathbb{E}[\theta_\ell] \geq \mathbb{E}[\theta_\ell^2]$, i.e. $\bar{\theta} \geq \mu_{\ell,2}/\mu_{\ell,1}$.

E.5 Estimates and Discussion

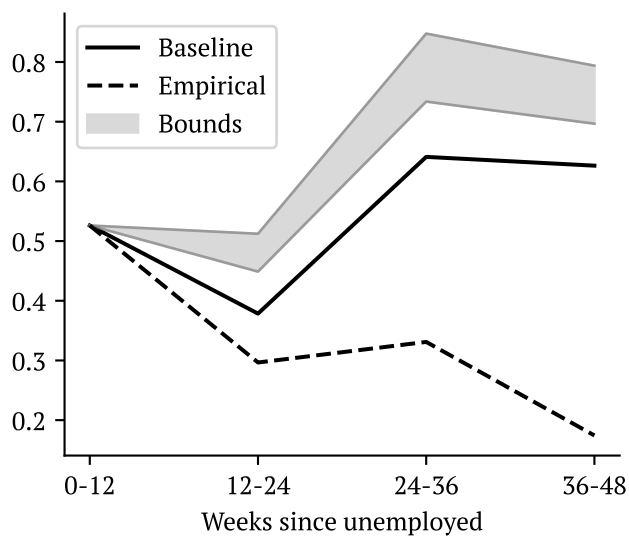
Finally, Figure E2 reports estimates of the structural hazard under the generalized model. The specification is identical to the baseline, except that I relax the restriction $\eta = 1$ and re-estimate the structural hazard at the two endpoints of the identified set for η . Because the estimates are monotonic in η , the resulting band contains the estimates corresponding to all intermediate values. Since the identified set lies above one, imposing $\eta = 1$ understates the dispersion of unobserved types. As a result, the interval of structural hazard estimates under the generalized model lies above the baseline estimates. Importantly, the qualitative pattern is unchanged: the structural hazard initially declines, increases around UI exhaustion, and remains relatively flat thereafter. I continue to report the baseline estimates based on $\eta = 1$ as the headline results in the main text because they provide a conservative benchmark, attributing less of the observed decline in job-finding rates to unobserved heterogeneity.

FIGURE E1: MOMENT INEQUALITIES BOUNDING η



Note: The panels show the restrictions that partially identify the composite parameter $\eta = \gamma\kappa$. Given η , the model is just-identified, so the moments of the implied type distribution are recovered as functions of η ; admissible values are those for which these moments are valid, i.e. correspond to a non-negative type distribution. Each panel plots one of the moment inequalities: panel (a) the second normalized moment $\lambda_2(\eta)$ against the variance bound of one; panel (b) the third moment $\lambda_3(\eta)$ against $\lambda_2(\eta)^2$; and panel (c) the fourth moment $\lambda_4(\eta)$ against the floor implied by positive semidefiniteness of the Hankel matrix. The upper bound $\bar{\eta} = 1.29$ follows from the bounded support of ν .

FIGURE E2: ESTIMATES IMPLIED BY BOUNDS

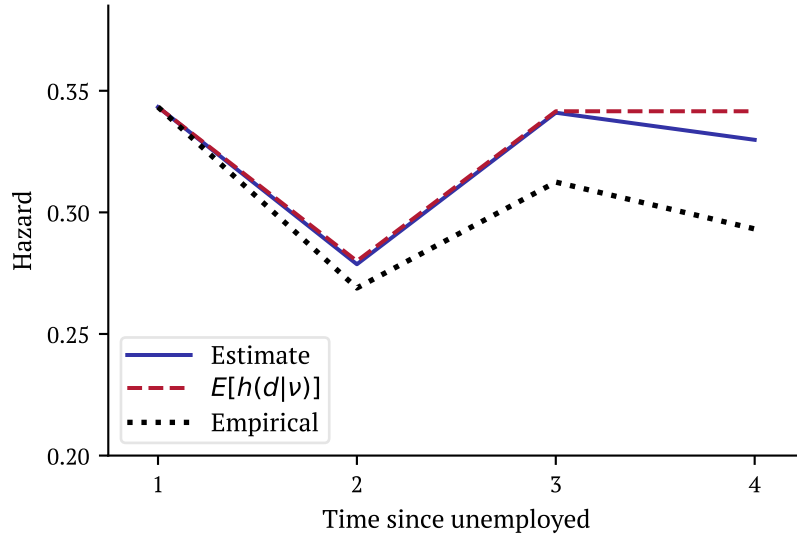


Note: The figure presents structural hazard estimates from the more generalized model in which the conditional independence and stationarity assumptions are relaxed, with the relaxation governed by a single composite parameter $\eta = \gamma\kappa$ that is partially identified by the requirement that the implied type distribution be valid. The shaded band spans the estimates between the bounds on η , compared to the baseline estimate (solid) and the empirical hazard in the data (dashed).

APPENDIX F SEARCH MODEL SIMULATION

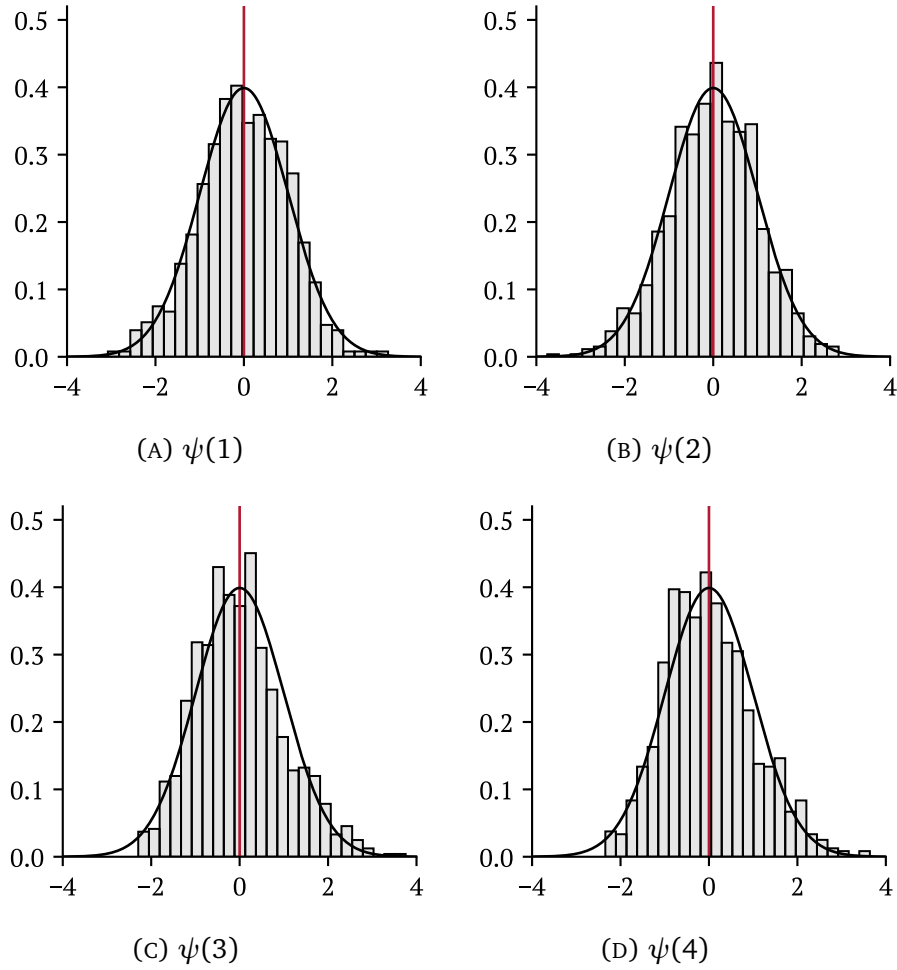
In this section, I simulate data from the search model presented in the main text. To incorporate multiple notice periods, I let the offer rate in the first period be different for long (L) and short (S) notice individuals. I set $\nu_H = 1$, $\nu_L = 0.5$ and $\pi = 0.5$, $\delta_L(1) = 1.25$, $\delta_S(1) = 1$, and $\delta(d) = 0.95$ for $d = 2, 3, 4$. The rest of the parameters are set as in the calibration of the model in the main text. I assume there are 3000 individuals, half of whom receive the L length notice. I simulate data on exit rates for this model 1000 times. The average of estimates for the structural hazard is presented in Figure F1, while the distribution of the estimates is presented in Figure F2.

FIGURE F1: SIMULATION: AVERAGE ESTIMATE



Note: The solid blue line presents the average estimate from 1000 simulations of the search model. The dashed red line presents the structural duration dependence $\mathbb{E}[h(d|\nu)]$ implied by the model. While the dotted black line presents the observed structural duration dependence $\mathbb{E}[h(d|\nu)|D \geq d]$ implied by the model.

FIGURE F2: ESTIMATES USING SIMULATED DATA FROM THE SEARCH MODEL



Note: The figure presents the normalized distribution of structural duration dependence estimated on simulated data from the search model. The vertical lines represent the mean of the distribution for each structural hazard. Standard normal density is overlaid for reference.